

Knowledge graph (KG)

COMPCSI 752: Big data management

University of Auckland

Slides are collected and edited from <https://web.stanford.edu/class/cs520/> (Galkin)
and <https://kgtutorial.github.io/> (WSDM 2018 Tutorial)
and <https://kdd2018tutorialt39.azurewebsites.net/> (KDD 2018 Tutorial)

Credit to Ninh Pham for these slides

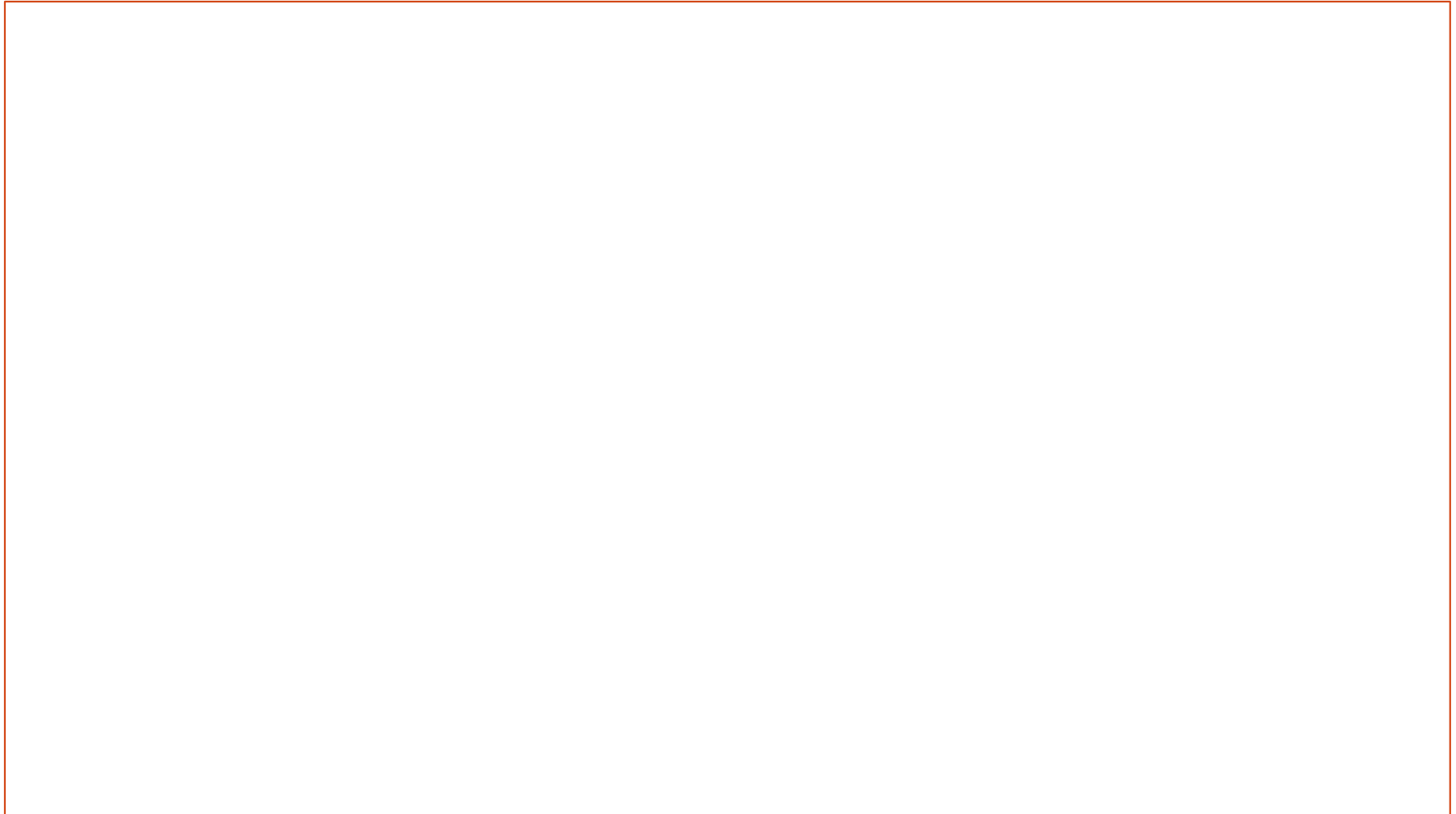
Outline

- Introduction of knowledge graph
- How to construct knowledge graph
- Challenges and success stories

Outline

- Introduction of knowledge graph
 - Introduction and applications
 - Viewpoints: Logic, Databases, CV, NLP
- How to construct knowledge graph
- Challenges and success stories

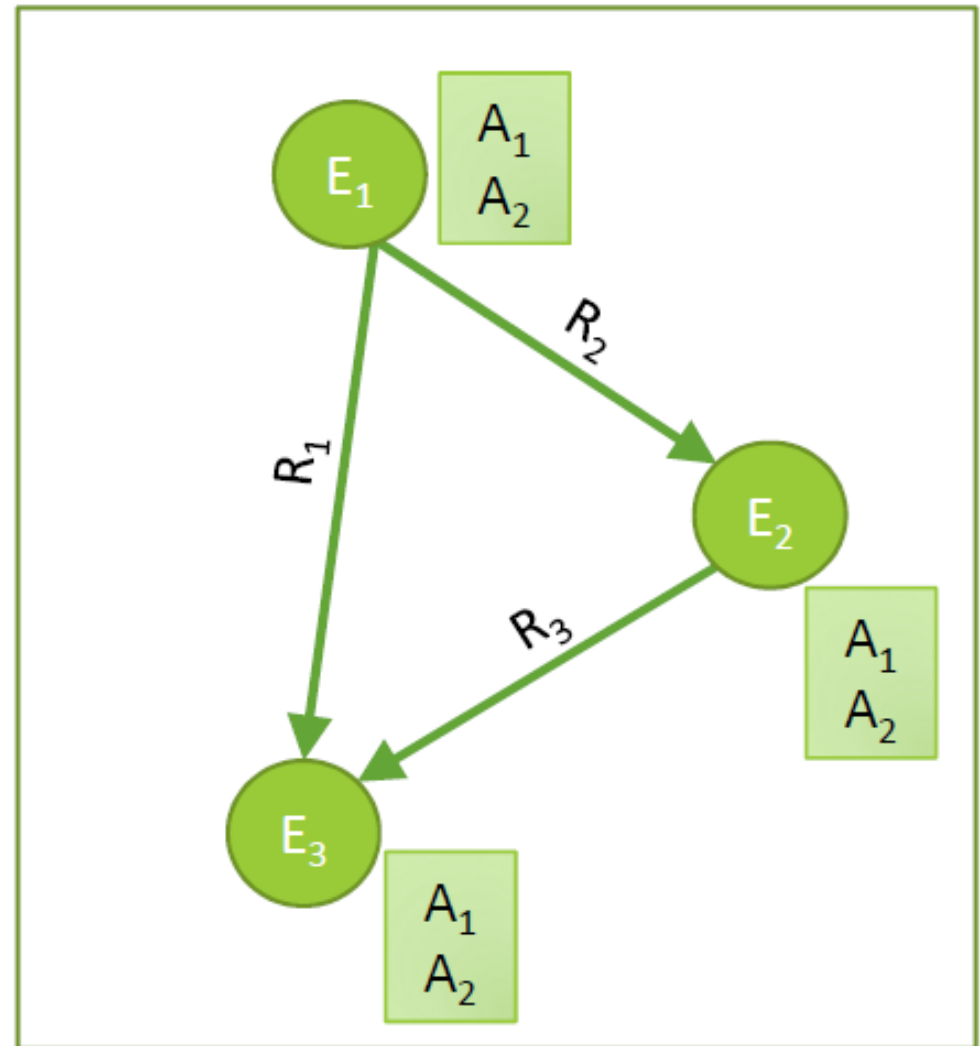
KG: Google's knowledge base



<https://www.youtube.com/watch?v=mmQl6VGvX-c&t=75s>

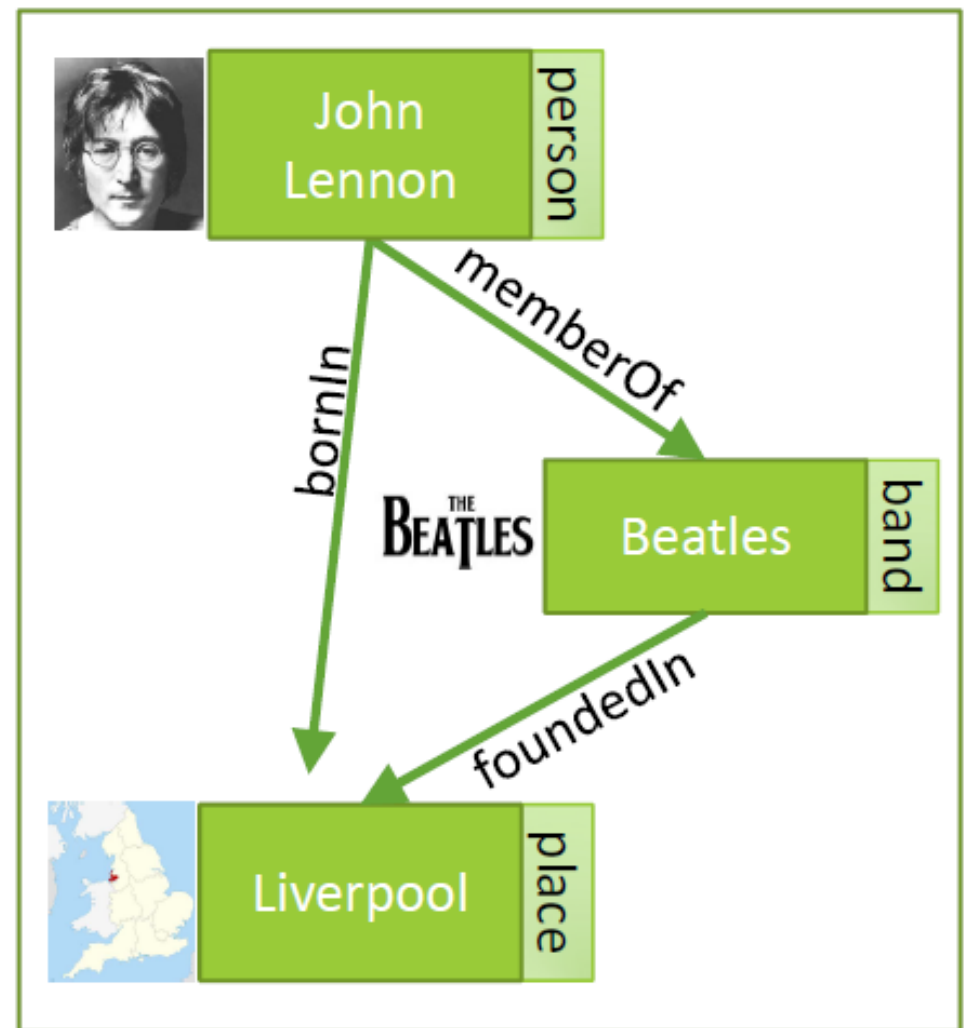
What is KG?

- Knowledge in graph form!
- Capture **entities**, **attributes** and **relationships**
- **Nodes** are entities
- Nodes are labeled with **attributes** (e.g. types)
- **Edges** between 2 nodes capture a relationship between entities



KG example

- Knowledge in graph form!
- Capture **entities**, **attributes** and **relationships**
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Why KGs?

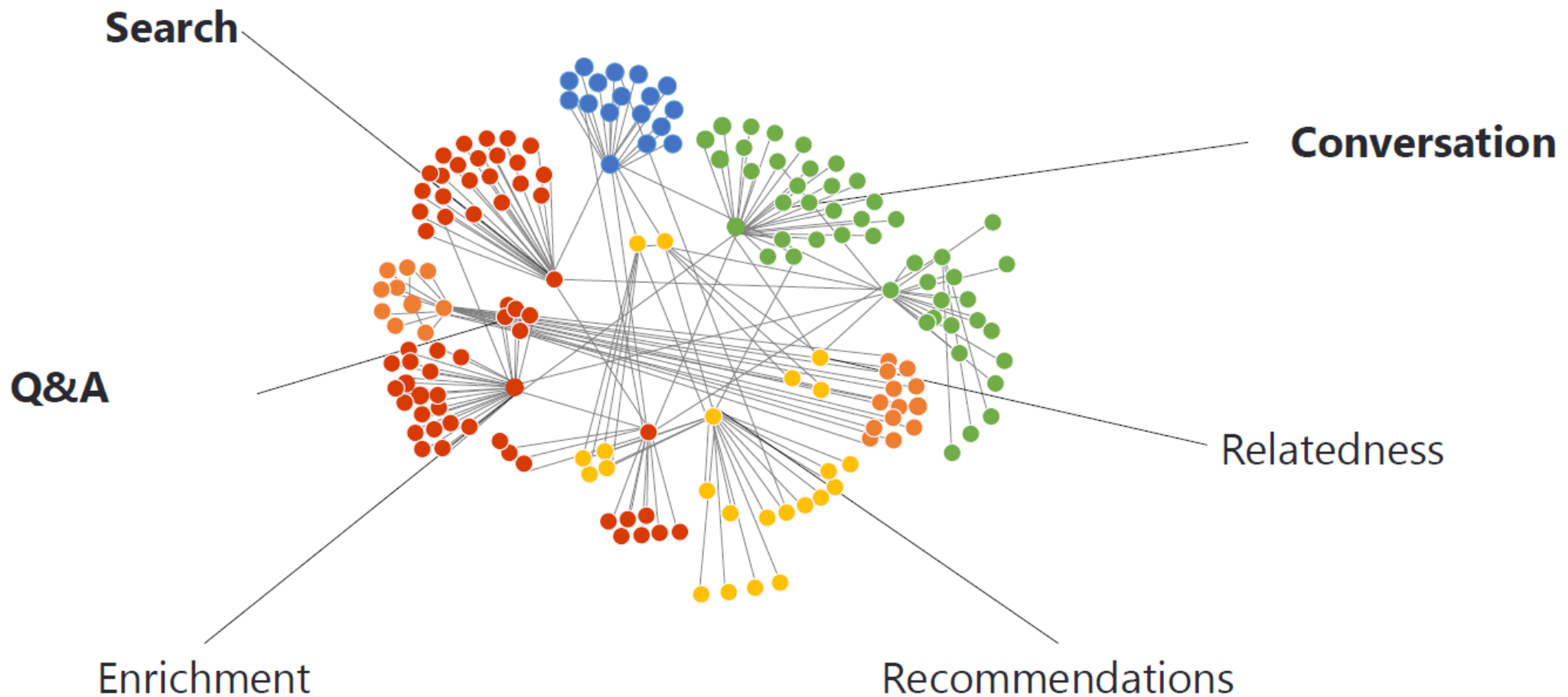
- Human:

- Combat information (from big data) overload
- Explore via intuitive structure
- Tool for supporting knowledge-driven tasks

- AIs:

- Key ingredient for many AI tasks
- Bridge from data to human semantics
- Use decades of work on graph analysis

Applications



Powering Google search

big bang theory

×





 All

 Images

 News

 Videos

 Maps

 More

Settings

Tools

About 161,000,000 results (0.85 seconds)

en.wikipedia.org › wiki › The_Big_Bang_Theory ▼

The Big Bang Theory - Wikipedia

The **Big Bang Theory** is an American television sitcom created by Chuck Lorre and Bill Prady, both of whom served as executive producers on the series, along with Steven Molaro. All three also served as head writers.

Related shows: [Young Sheldon](#) **Producer(s):** [Faye Oshima Belyeu](#)

No. of seasons: 12 **Original release:** September 24, 2007 –; May 1...

[List of The Big Bang Theory ...](#) · [Sheldon Cooper](#) · [Category:The Big Bang Theory](#)

www.imdb.com › title ▼

The Big Bang Theory (TV Series 2007–2019) - IMDb

★★★★★ Rating: 8.1/10 - 684,017 votes

Created by Chuck Lorre, Bill Prady. With Johnny Galecki, Jim Parsons, Kaley Cuoco, Simon Helberg. A woman who moves into an apartment across the hall ...

People also ask

What is Big Bang theory of universe?

▼

Why did Big Bang Theory end?

▼

Is Big Bang Theory on a break?

▼

Is Big Bang Theory on Netflix?

▼

Feedback



The Big Bang Theory

2007 · Sitcom · 12 seasons

8.1/10

IMDb

82%

Rotten Tomatoes

8.9/10

TV.com

94% liked this TV show

Google users



The lives of four socially awkward friends, Leonard, Sheldon, Howard and Raj, take a wild turn when they meet the beautiful and free-spirited Penny.

First episode date: 24 September 2007

Final episode date: 16 May 2019

No. of episodes: 279 (list of episodes)

Knowledge-powered Q&A

who is vice chancellor of the university of auckland



All Images News Videos Shopping Web Forums More

Tools

AI Overview

Learn more

The current Vice-Chancellor of the University of Auckland is **Professor Dawn Freshwater**.

Role:

She is the university's chief academic and administrative officer, and the employer of all staff.

Responsibilities:

She is responsible for academic leadership, effective management, strategic planning, and resource allocation.

Office Location:

Show more



Management team - The University of Auckland

The University of Auckland

Vice-Chancellor - The University of Auckland

Professor Dawn Freshwater, BA(Hons) Manc., PhD Nott.

The University of Auckland

Law school stoush: Auckland Uni vice-chancellor faces backlash

14 Mar 2020 - University of Auckland vice-chancellor Dawn

Professor Dawn Freshwater

Professor Dawn Freshwater, BA(Hons) Manc., PhD Nott.



The University of Auckland

<https://www.auckland.ac.nz/about-us/the-university>

Vice-Chancellor - The University of Auckland

About featured snippets • Feedback

Knowledge-powered conversation

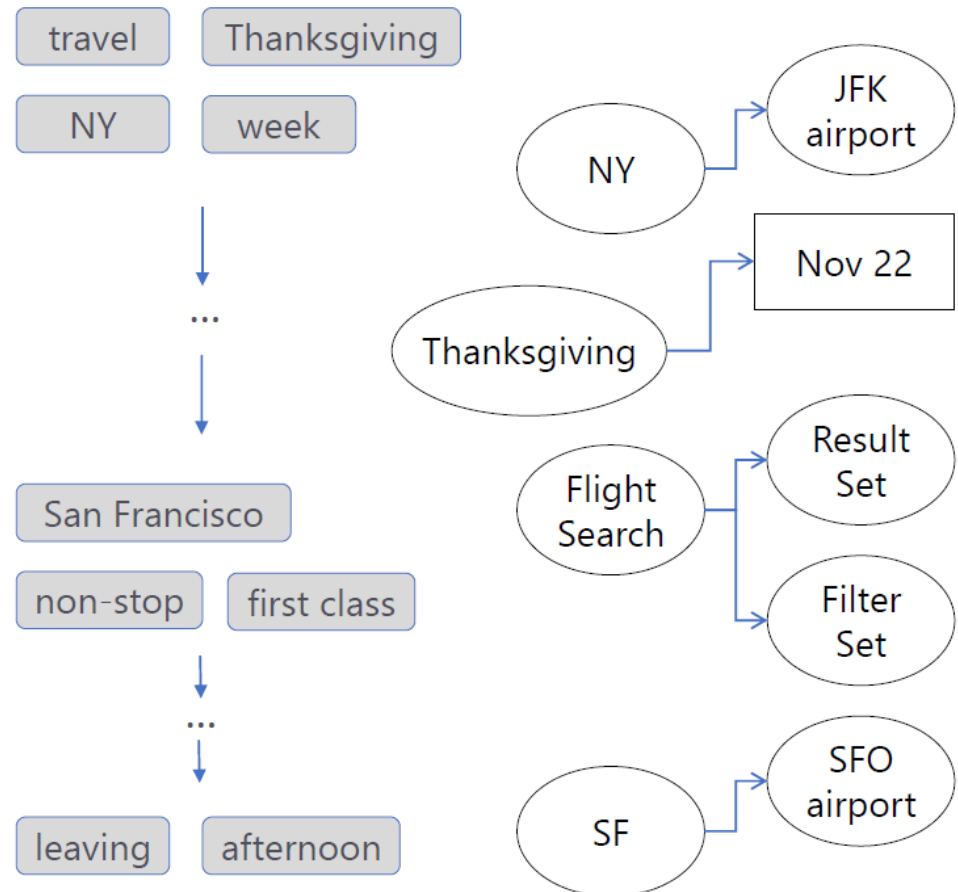
I want to travel to NY 2 days before Thanksgiving, staying for a week

Okay, booking a flight to JFK from November 20 to November 27. Where will you be flying from?

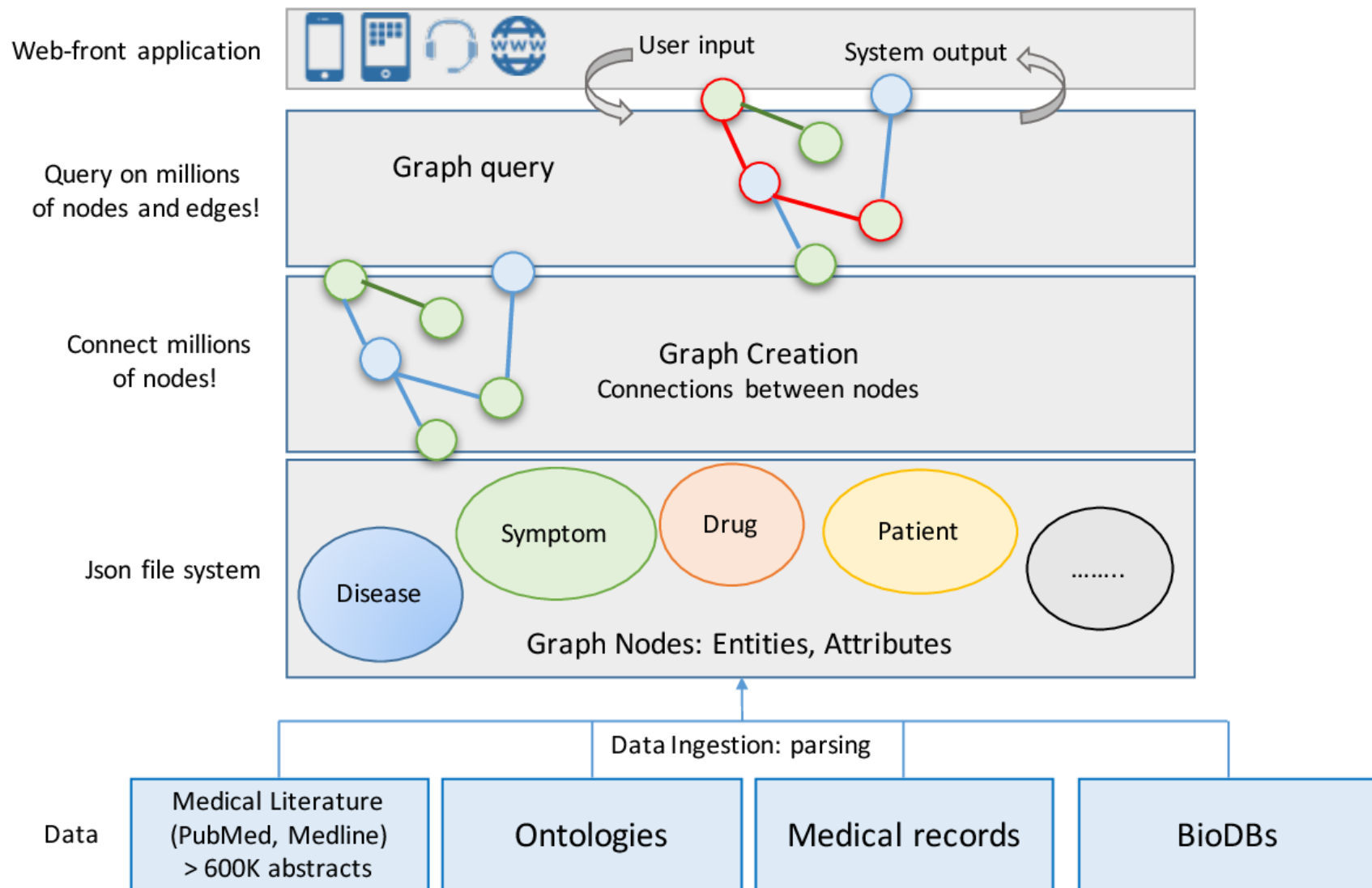
From San Francisco, and also non-stop in first class

Got it, I've found some flights for you ...

How about leaving in the afternoon



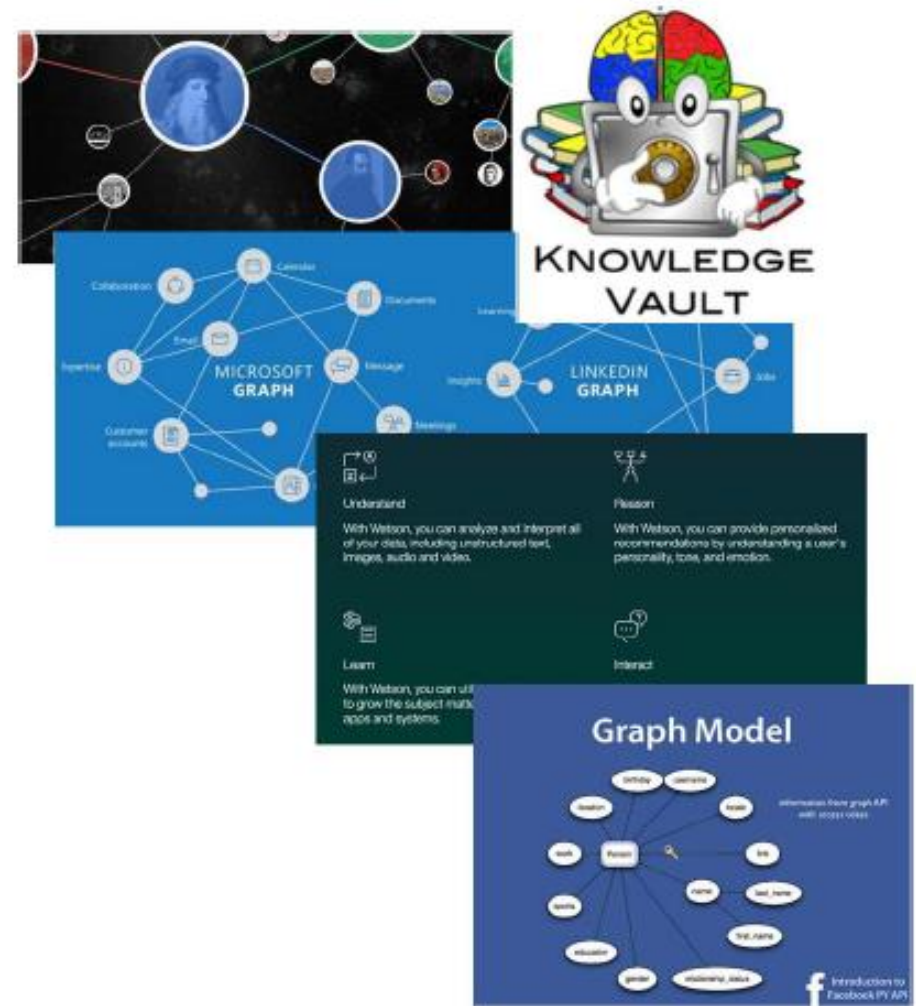
Decision support



Source: Cognitive DDx Assistant in Rare Diseases (Reumann et al. 2018)

KG and industry

- Google KG
- Amazon Product Graph
- Facebook Graph API
- IBM Watson
- Microsoft Satori
- LinkedIn KG
- Yandex Object Answer
- ...



KGs meet LLMs

Knowledge Graphs Status Quo

Has the network been infected by the Trickbot malware?

- Long turnaround times
- Added burden to expert team
- Hard to ask new/emergent questions

Business Analyst

Expert Team

Network of 100 nodes (Business Analyst, Trickbot, Malware, etc.)
- Trickbot (malware) is a threat to the network.
- Trickbot (malware) is a threat to the network.
- Trickbot (malware) is a threat to the network.
- Trickbot (malware) is a threat to the network.

Where does knowledge come from?

- Structured text:
 - Wikipedia, tables, databases

The Beatles				
				
The "Fab Four" Beatles lineup in 1964. Clockwise from top left: John Lennon, Paul McCartney, Ringo Starr and George Harrison				
Background Information				
Origin	Liverpool, England, United Kingdom			
Genres	Rock • pop			
Years active	1960–1970			
Labels	EMI • PolyGram • Swan • Vee-Jay • United Artists			
Associated acts	The Quarrymen • The Quarrymen • Preston • The Beatles			
Website	thebeatles.			
Past members	John Lennon • Paul McCartney • George Harrison • Ringo Starr			

© National Oceanography Centre, Liverpool				
Mon 30th Jan 2017	00:18 9.15m H	07:06 1.34m L	12:36 9.50m H	19:32 1.20m L
Tue 31st	00:55 9.18m H	07:43 1.36m L	13:14 9.49m H	20:10 1.25m L
Wed 1st Feb 2017	01:33 9.10m H	08:21 1.51m L	13:53 9.37m H	20:47 1.42m L
Thu 2nd	02:14 8.91m H	08:59 1.78m L	14:36 9.15m H	21:27 1.70m L
Fri 3rd	03:00 8.63m H	09:42 2.08m L	15:24 8.84m H	22:12 2.04m L
Sat 4th	03:52 8.27m H	10:34 2.43m L	16:21 8.45m H	23:09 2.39m L
Sun 5th	04:59 7.95m H	11:42 2.71m L	17:34 8.13m H	
Mon 6th	00:24 2.63m L	05:20 7.82m H	13:09 2.73m L	18:57 8.06m H
Tue 7th	01:49 2.56m L	07:39 8.03m H	14:31 2.42m L	20:13 8.29m H
Wed 8th	03:03 2.23m L	08:49 8.46m H	15:43 1.93m L	21:18 8.69m H
Thu 9th	04:08 1.82m L	09:47 8.94m H	16:45 1.41m L	22:14 9.07m H
Fri 10th	05:03 1.44m L	10:38 9.34m H	17:38 0.99m L	23:01 9.35m H
Sat 11th	05:51 1.17m L	11:21 9.61m H	18:24 0.75m L	23:44 9.47m H

The Beatles Total Album Sales Statistics		Date
Total number of Beatles albums sold		2,303,100,000
Total Albums Sold on iTunes		765,000
Total Singles Sold on iTunes		9,805,000
Sales by Available Markets		
United States		309.1 Million
Canada		13.6 Million
United Kingdom		7.5 Million
Germany		7.3 Million
France		9.1 Million
Australia		2.8 Million
Japan		1.9 Million
Argentina		1.6 Million
Brazil		606,000
Sweden		564,000
Austria		575,000
Switzerland		468,000

Beatles Billboard Chart Statistics	
Total weeks on chart	1,278 weeks
Total number ones	16
Total weeks at number one	175 weeks
Albums with longest time spent at number one (Please Please Me)	30 weeks

© FREE DATA SET

NEXT DATA SET

Where does knowledge come from?

- Structured text:
 - Wikipedia, tables, databases
- Unstructured text:
 - WWW, news, social media

Beatles last live performance

Published: Thursday, January 26th 2017, 5:24 am PST

Updated: Monday, January 30th 2017, 4:06 am PST

Written by Jim Eftink, Producer [CONNECT](#)



(KPV5) - How about a little Beatles history.

It was on this date in 1969, the band performed their last live public performance.

Allan Williams, First Manager of the Beatles, Dies at 86

(Source: Stock int) By ALLAN KOZINN DEC. 31, 2016

The Beatles
January 17 at 10:00am · 45

The Harrison family is proud to announce the release of George Harrison - The Vinyl Collection box set featuring all of George Harrison's solo studio albums in one collection for the first time.

GEORGE HARRISON - THE VINYL COLLECTION
Released on 24th February, 2017, the vinyl box set includes all twelve of George's studio albums with exact replicas of the original release track listing and artwork. Also included in the box set are George's classic live album Live in Japan (2x) ... See More

George Harrison - The Vinyl Collection - Released February 24th 2017
George Harrison - The Vinyl Collection, available to pre-order now with an exclusive & limited edition...
YOUTUBE.COM

Like Comment All Share

Who knew you'd be...

Top Comments

888 shares

Write a comment...

Jeffrey Smith What I would really be interested in is an '80s thing, Must Pass... 'Shrieked Down' with just the basic tracks without Phil Spector's Wall of Sound. It's not 2 would sound really good and I would buy it in a heartbeat. Like · Reply · play · January 17 at 10:20am

17 Replies

Dave Standing I can just see the greedy Harrison family and the greedy music industry multi-millionaire big wigs rubbing their hands with glee once more while discussing various methods to make people buy their already bought and paid for record collections all at once. See More
Like · Reply · play · January 17 at 10:15am · Edited

30 Replies

View more comments 2 of 180

The Beatles
January 17 at 9:58am · 45

"Of very few individual songs can it be said, 'This changed the course of popular music.' 'A Day in the Life' is one such song." - Richard Havers

The Beatles - A Day in the Life
A Day in the Life The Beatles' Video Collection is Out Now. Get your copy here!



manager of the Beatles in 1960, he sent them on a stint in Germany tagcraft. Press Association, via Associated Press

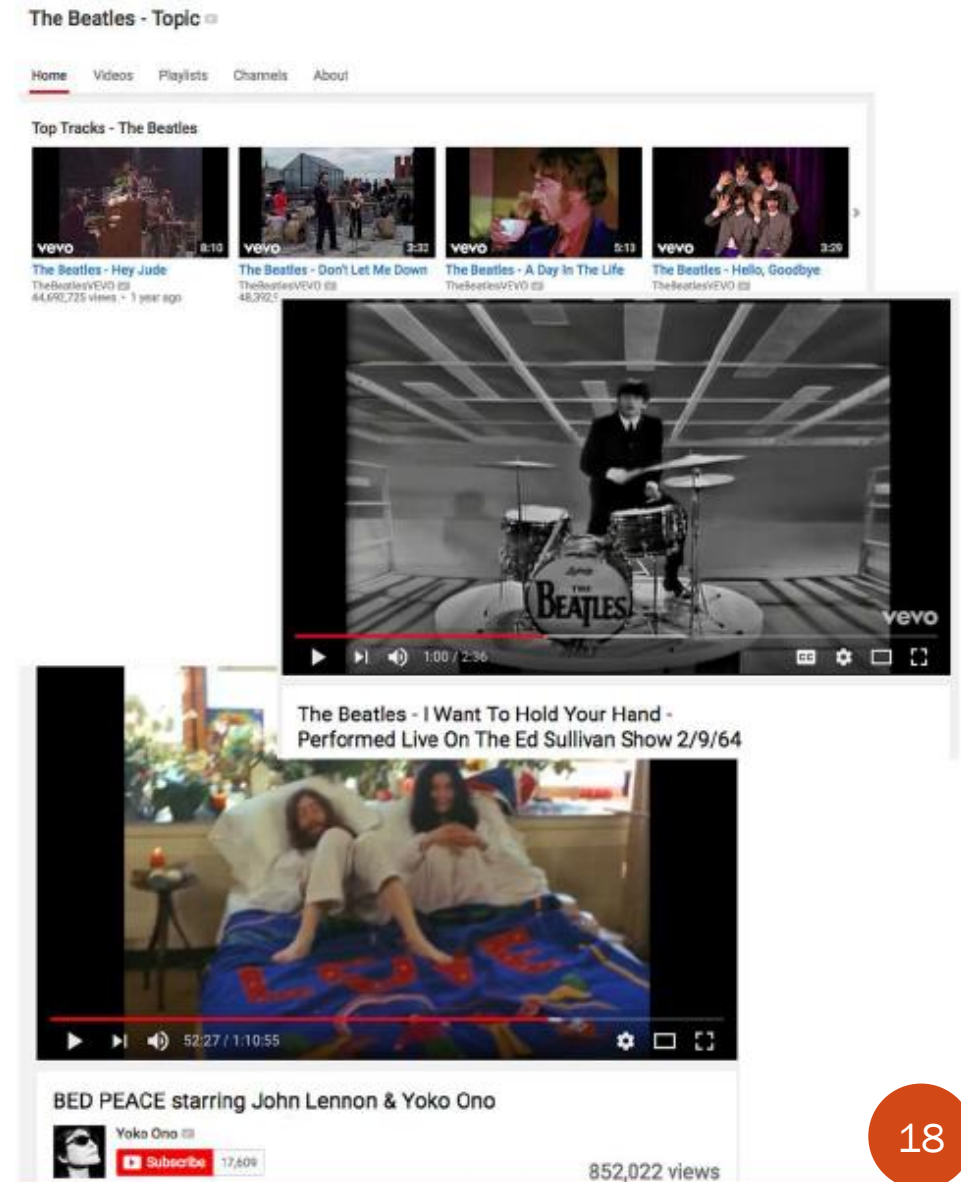
Where does knowledge come from?

- Structured text:
 - Wikipedia, tables, databases
- Unstructured text:
 - WWW, news, social media
- Images



Where does knowledge come from?

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- Video:
 - Youtube, video feeds



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- Introduction of knowledge graph
 - Introduction and applications
 - Viewpoints: Logic, Databases, CV, NLP
- How to construct knowledge graph
- Application of knowledge graph

Knowledge representation

Given entities $\mathcal{E}$, relations $\mathcal{R}$, KG is a directed multi-relational graph $\mathcal{G}$ that comprises triples (s, p, o)

$$\mathcal{G} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$$
$$(s, p, o) \in \mathcal{G}$$

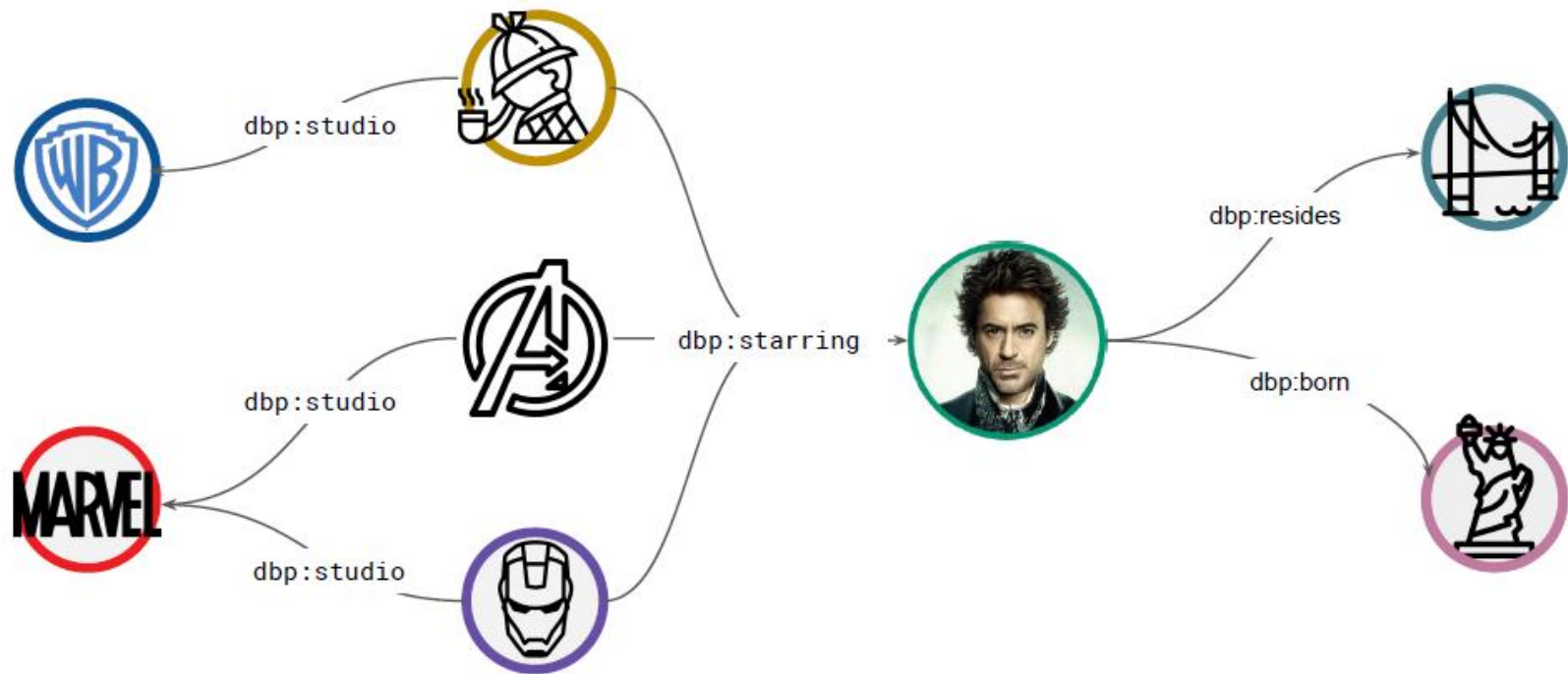
- * describes entities and relations
- * defines a schema
- * interrelating arbitrary entities
- * various topical domains

“Schema (TBox) with instances (ABox)”

“Every RDF / LPG / RDF* graph is a knowledge graph”

Graph-structured world model

Symbolic: Triples



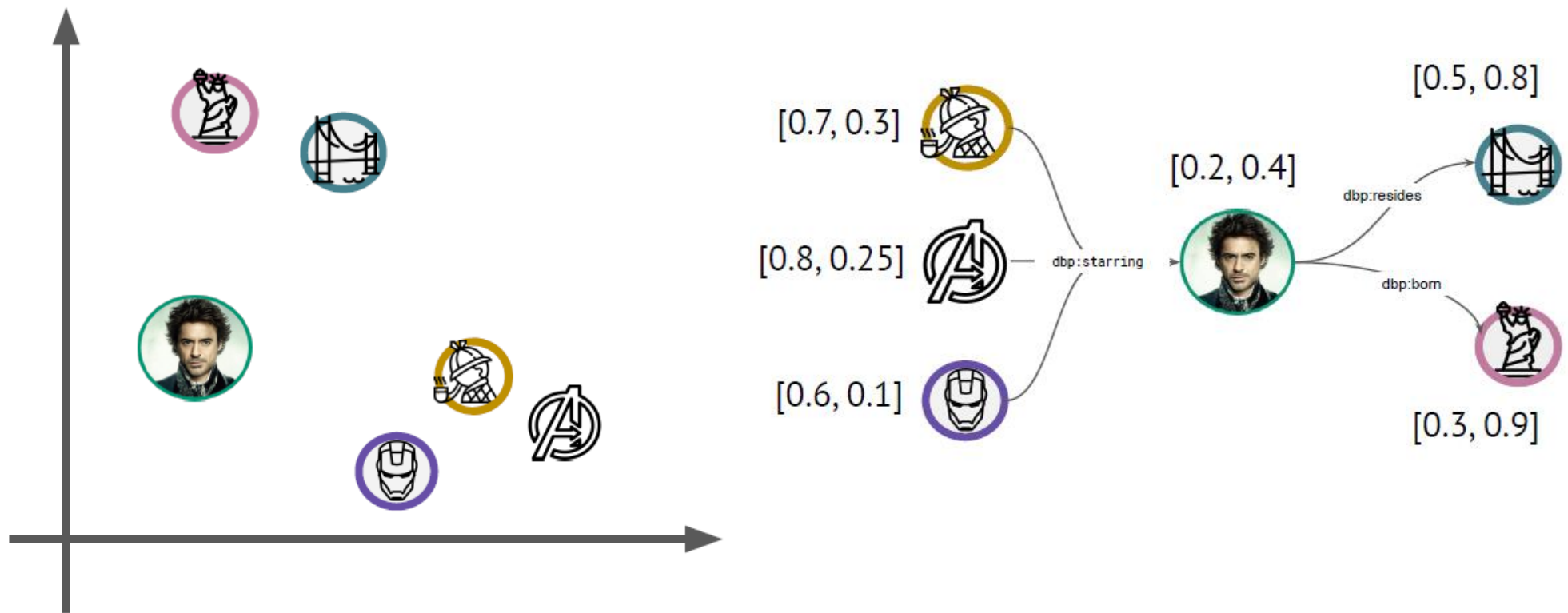
RDJ
RDJ
Sherlock_Holmes
Sherlock_Holmes

dbp:resides SF .
dbp:born NY .
dbp:studio WB .
dbp:starring RDJ .

Avengers
Avengers
Iron_Man
Iron_Man

dbp:studio Marvel .
dbp:starring RDJ .
dbp:studio Marvel .
dbp:starring RDJ .

Vector: Embeddings



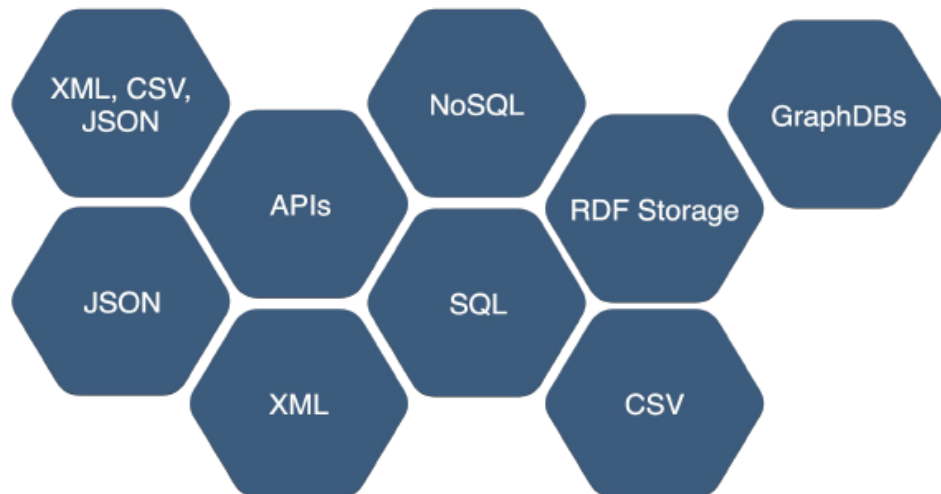
<http://rdf2vec.org/>

View: Database – Data integration

Knowledge Graph

Semantic Data Integration (Physical / Virtual)

Structured Sources



Entities

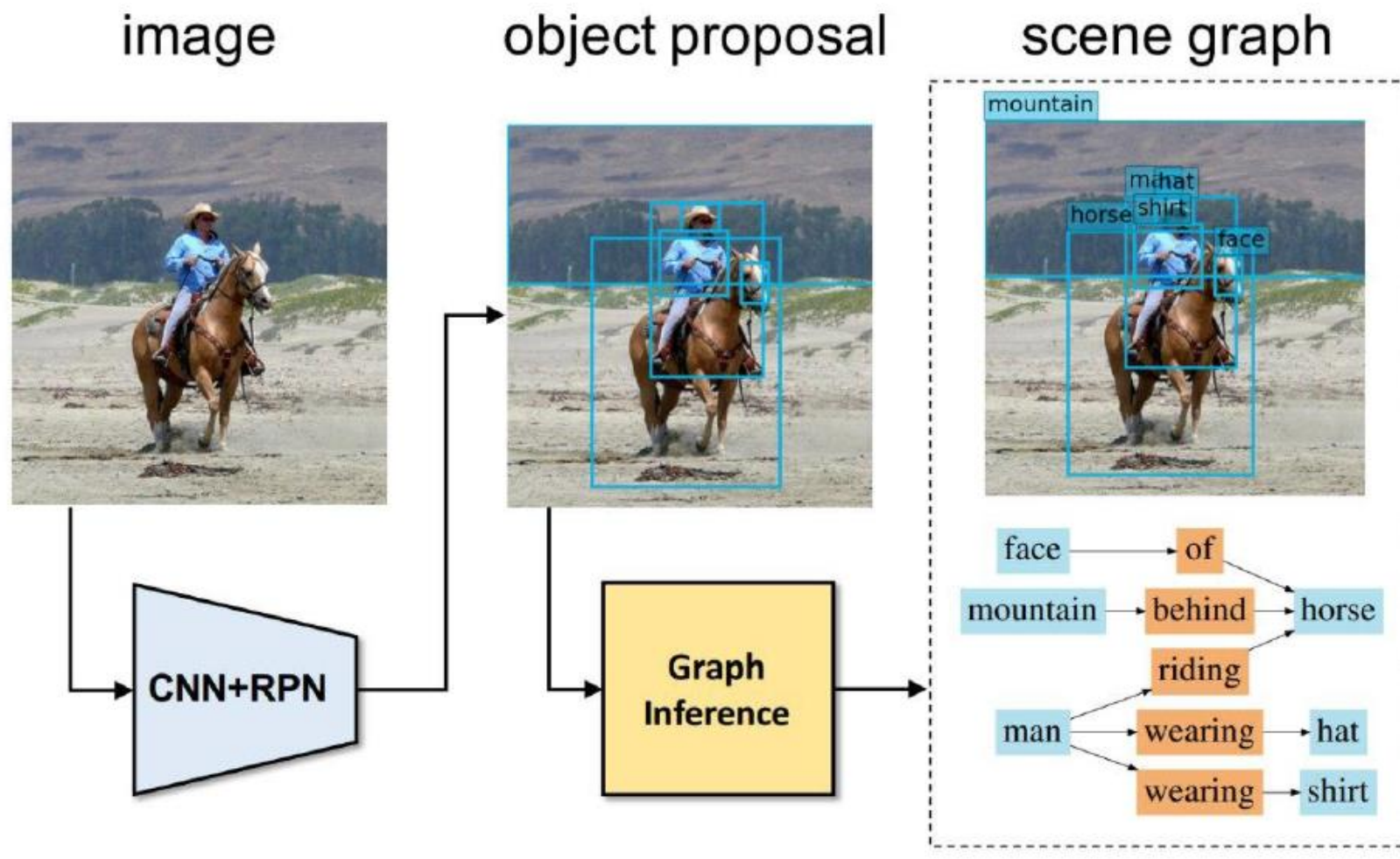
Cells - **classes**, **instances**

Relations

Column headers

Author	Notable work	Date of birth
Stan Lee	Iron Man	12/28/1922
Bob Kane	Batman	10/24/1915

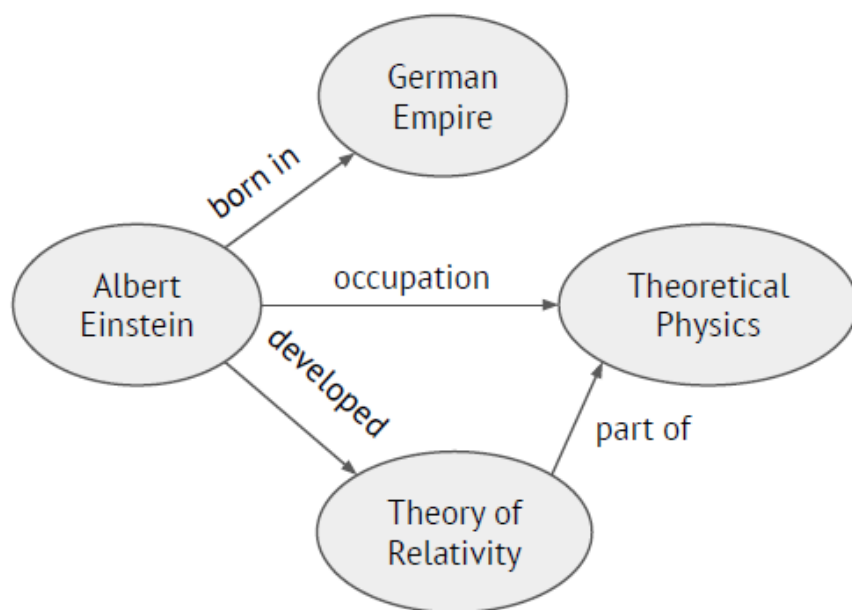
View: Computer vision



Xu et al. Scene Graph Generation by Iterative Message Passing. CVPR 2017

View: NLP – Build KGs from texts

Albert Einstein was a German-born theoretical physicist who developed the theory of relativity.



Knowledge Graph

Information Retrieval

Unstructured Sources



View: NLP – Named entity recognition

apple (Q89)

fruit of the apple tree
apples

Apple (Q1754545)

1990 album by Mother Love bone

Apple (Q213710)

UK international record label; imprint of Apple Corps Ltd.
LC 01074 | LC 1074 | Apple Records

Apple Inc. (Q312)

American producer of hardware, software, and services, based in Cupertino, California
Apple Computer, Inc. | Apple Computer | Apple Computer Inc | Apple | Apple Incorporated | Apple Computer Incorporated | 🍏

Who is the CEO of **Apple**?

 **Apple** belongs to which genus?

 **Downey** played **Iron Man** in which year?

Who is the alter ego of **Iron man**?

View: NLP – Relation linking

Relations in a Knowledge Graph

List of known relations

Surface forms (synonyms),
easily multi-lingual

Relations constraints

Relations hierarchy

Most used types of
subjects and objects

Name all the movies in which Robert Downey Jr ^{wdt:P161}**acted**?

Find me all the films **casting** Robert Downey Jr ?

List all the movies **starring** Robert Downey Junior?

RDJ **has acted** in which movies?

cast member (P161)

actor in the subject production |

starring | film starring | actor | actress | contestant or a play

performer (P175)

actor, musician, band or other performer associated with this role or musical work

artist | musician | played by | portrayed by | recorded by | recording by | dancer | actor | musical artist

All
marvel
movies

Every
thing
starring
RDJ

Find the
intersection

Count the
entities
left

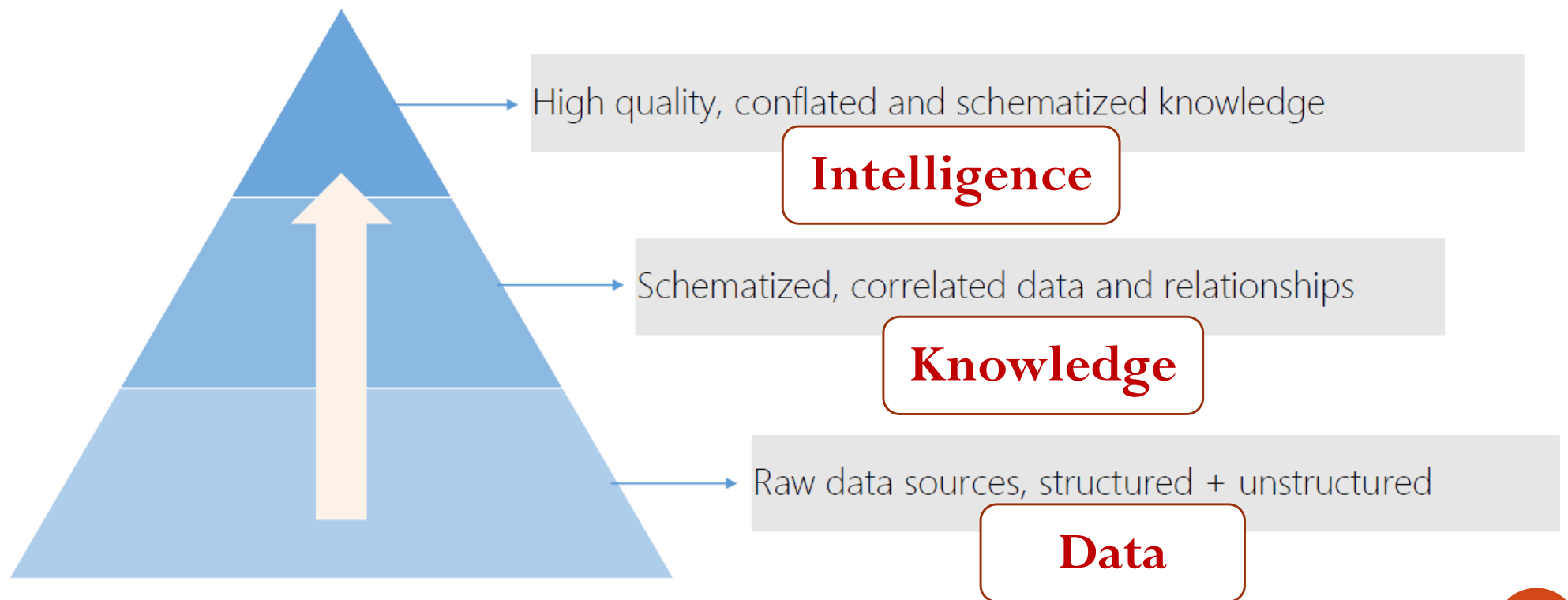
View: NLP – Q&A

How many **Marvel** **movies** was **Robert Downey Jr.**
casted in?

```
SELECT COUNT(?uri) WHERE {  
  ?uri dbp:studio dbr:Marvel_Studios.  
  ?uri dbo:starring dbr:Robert_Downey_Jr  
}
```

Create high-quality KGs

- Require machine learning, natural language processing, conflation and inference

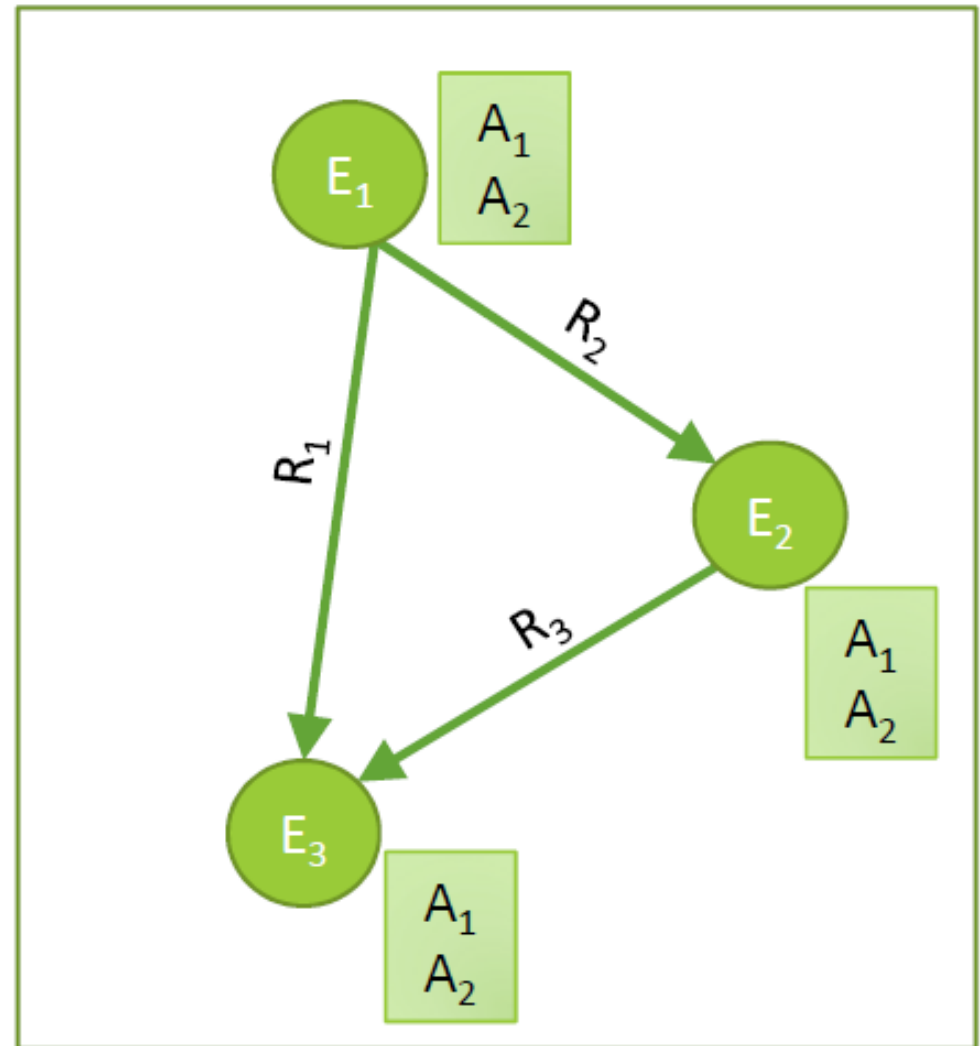


Outline

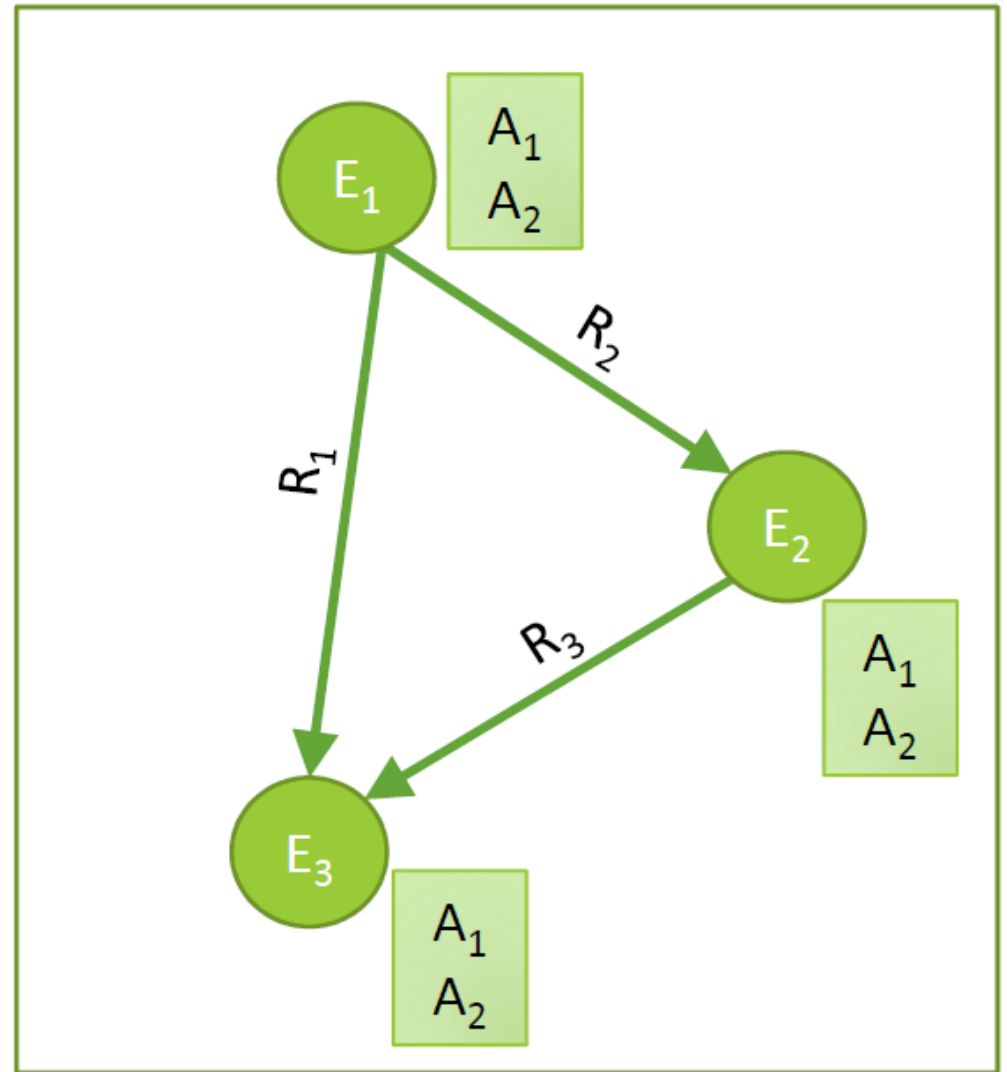
- Introduction of knowledge graph
- How to construct knowledge graph
 - Extract information (from text)
 - Match/merge entities
 - Knowledge refinement
- Challenges and success stories

What is KG?

- Knowledge in graph form!
- Capture **entities**, **attributes** and **relationships**
- **Nodes** are entities
- Nodes are labeled with **attributes** (e.g. types)
- **Edges** between 2 nodes capture a relationship between entities

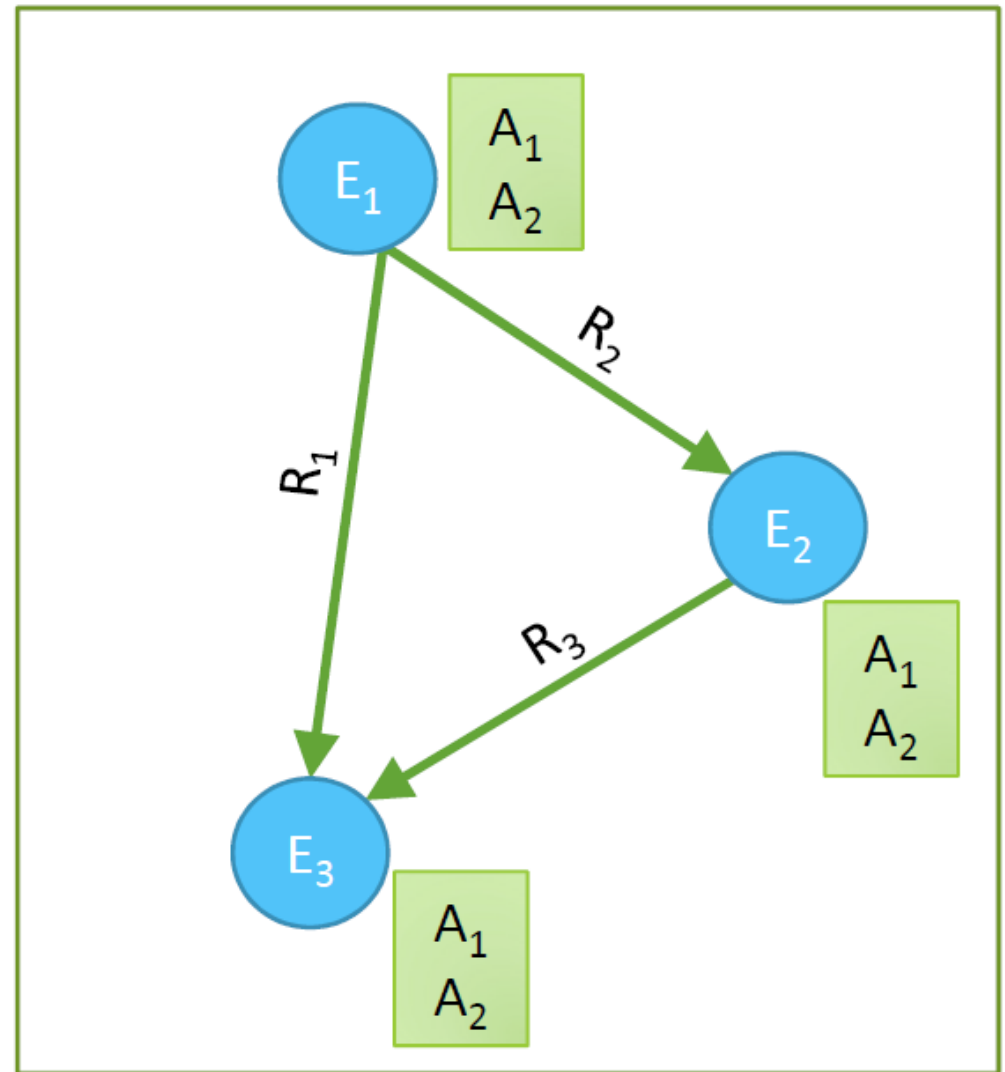


Basic problems



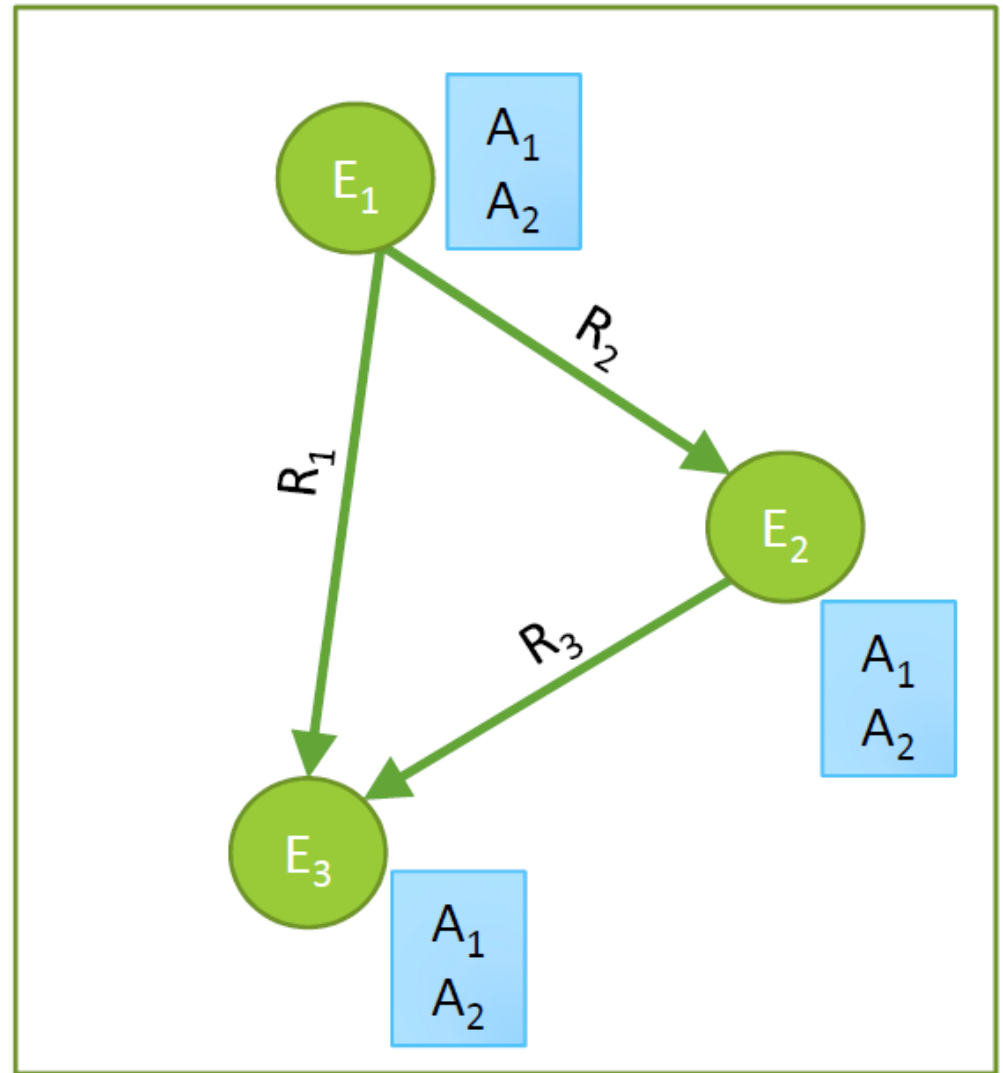
Basic problems

- Who are the entities (nodes) in graph?



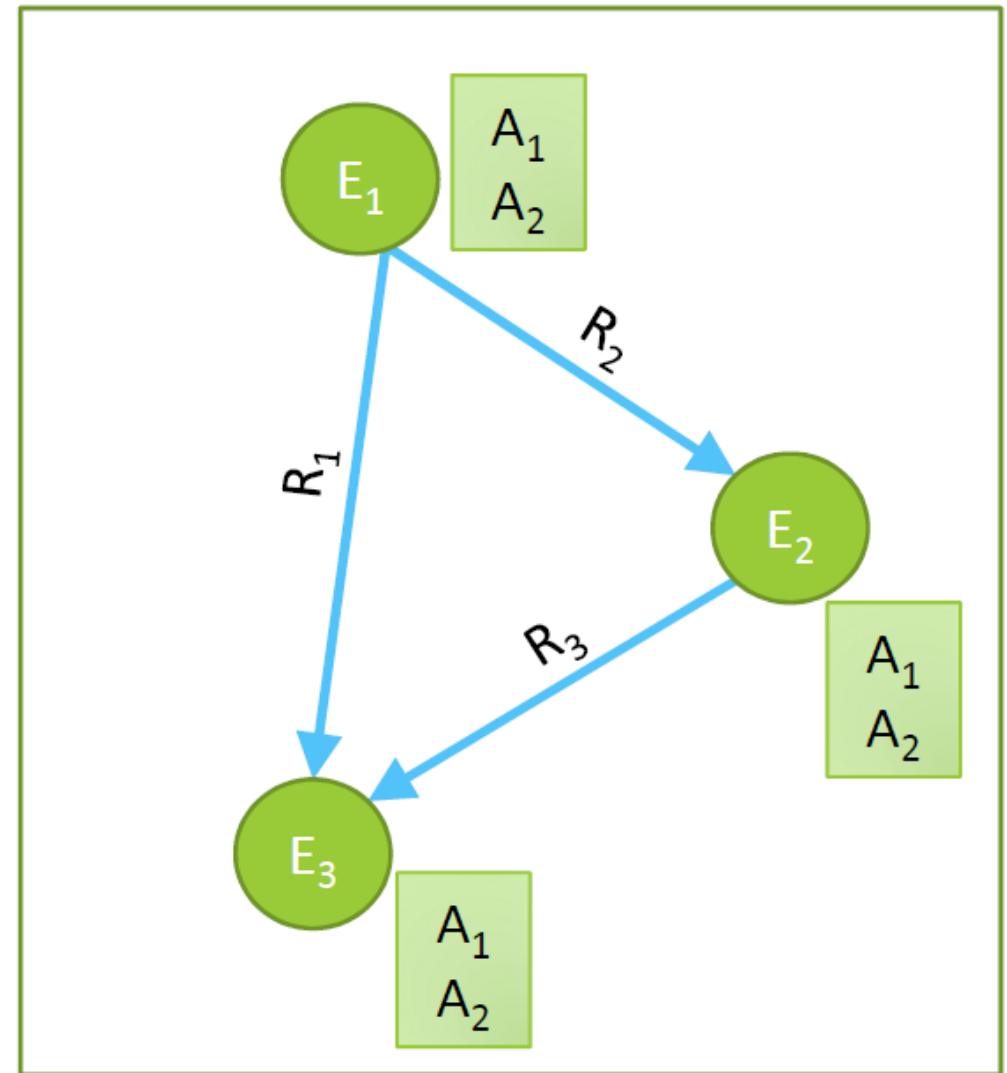
Basic problems

- Who are the entities (nodes) in graph?
- What are their attributes and types (labels)?



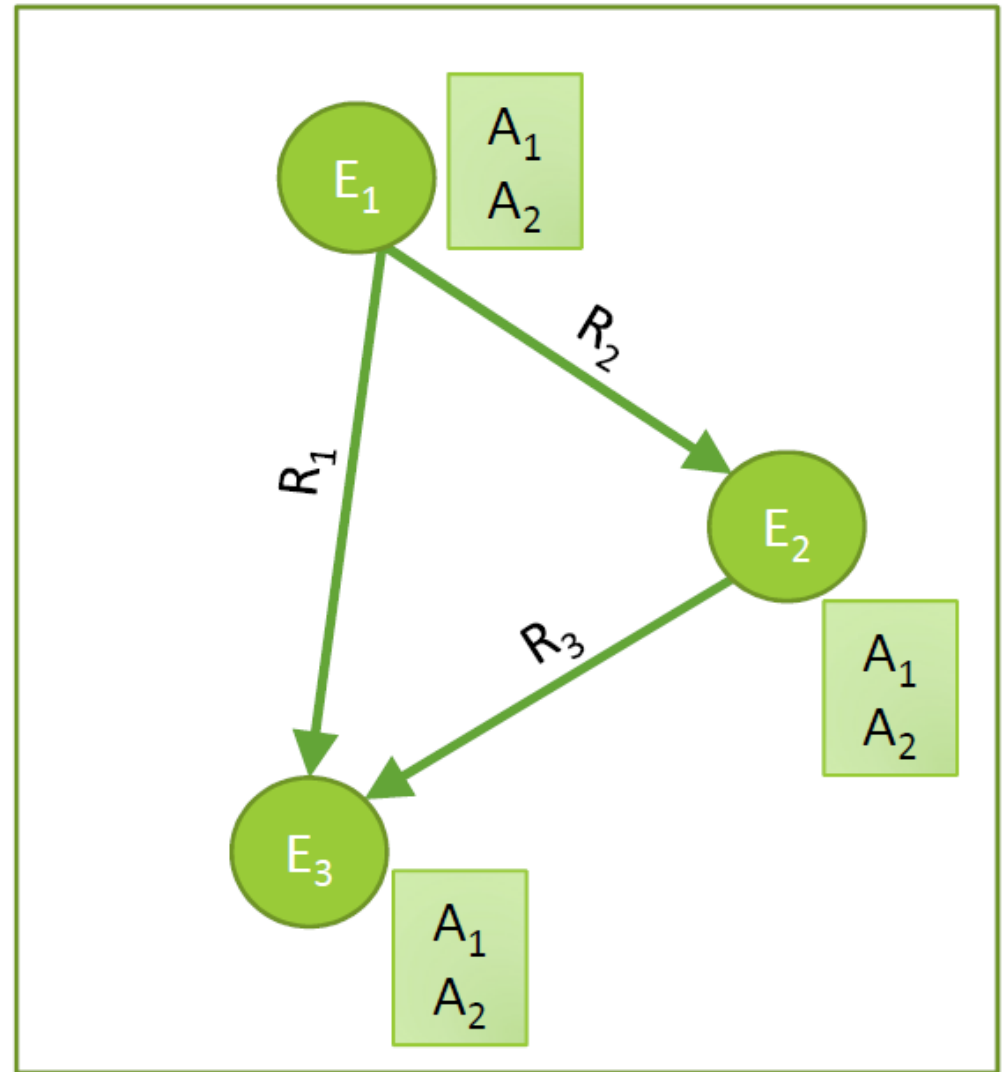
Basic problems

- Who are the entities (nodes) in graph?
- What are their attributes and types (labels)?
- **How** are they related (edges)?



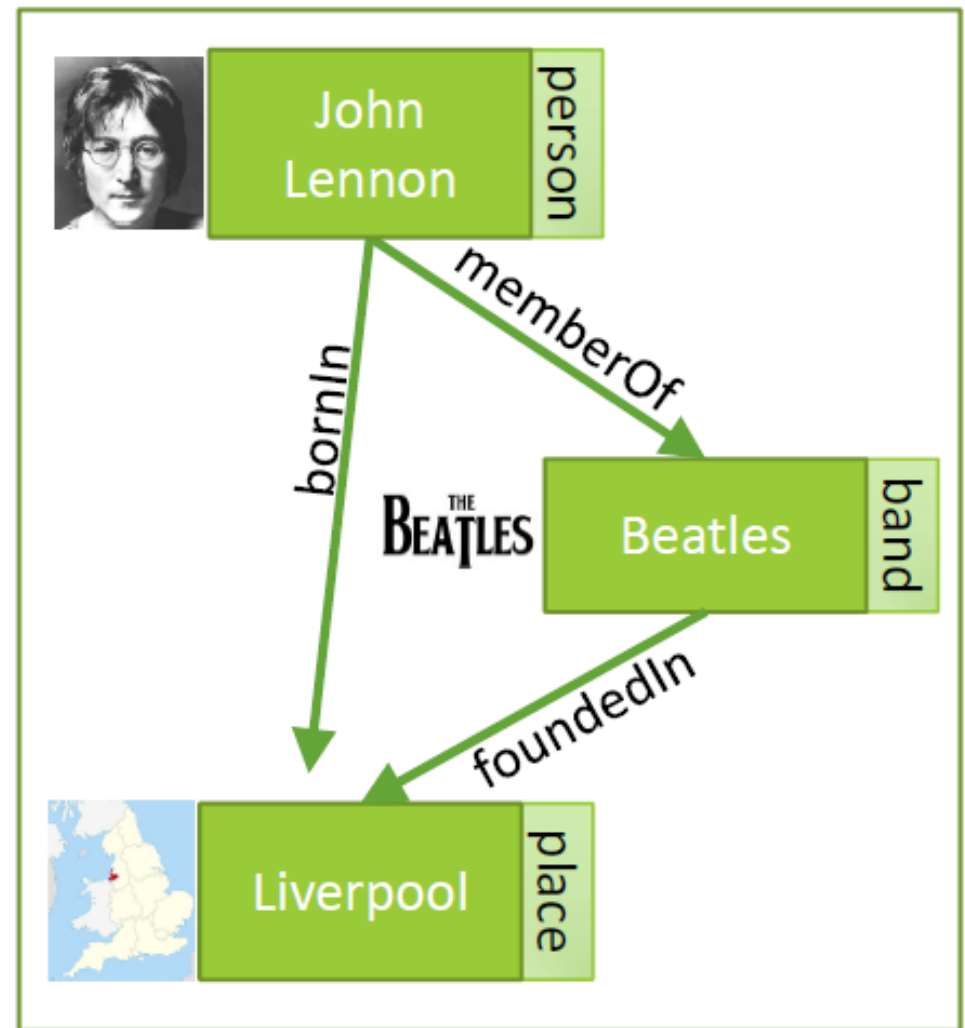
Basic problems

- **Who** are the entities (nodes) in graph?
- **What** are their attributes and types (labels)?
- **How** are they related (edges)?

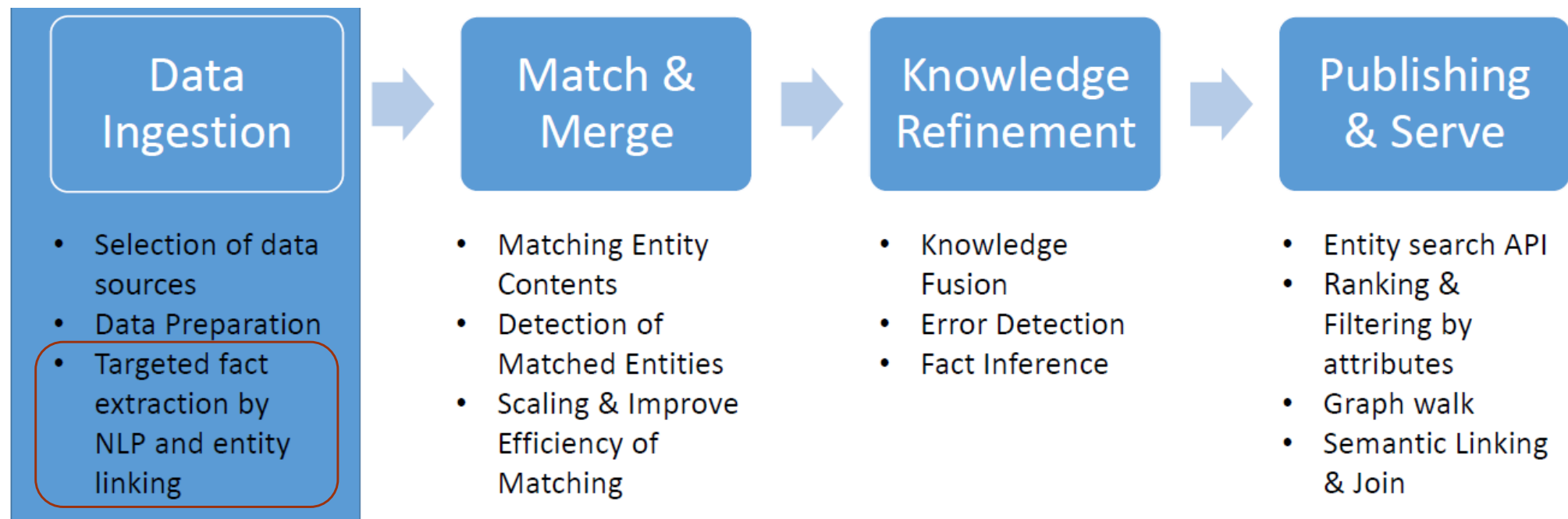


How difficult it is

- Easiest: Only Who
- Easy: Who + How
- Difficult: Who + How + What
- Very difficult: Who + How + What + Constraints



Construct KG

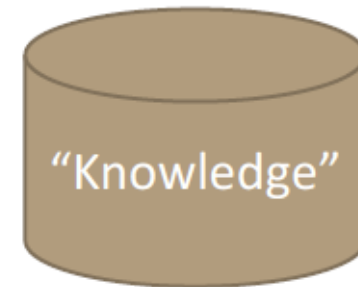


Knowledge extraction

Knowledge extraction from text



Unstructured
Ambiguous
Lots and lots of it!



Structured
Precise, Actionable
Specific to the task

Humans can read them, but
... very slowly
... can't remember all
... can't answer questions

Can be used for downstream
applications, such as creating
Knowledge Graphs!

Knowledge extraction using NLP

John was born in Liverpool, to Julia and Alfred Lennon.

Text

NLP

Lennon..
John Lennon...

the Pool

Mrs. Lennon..
.. his mother ..

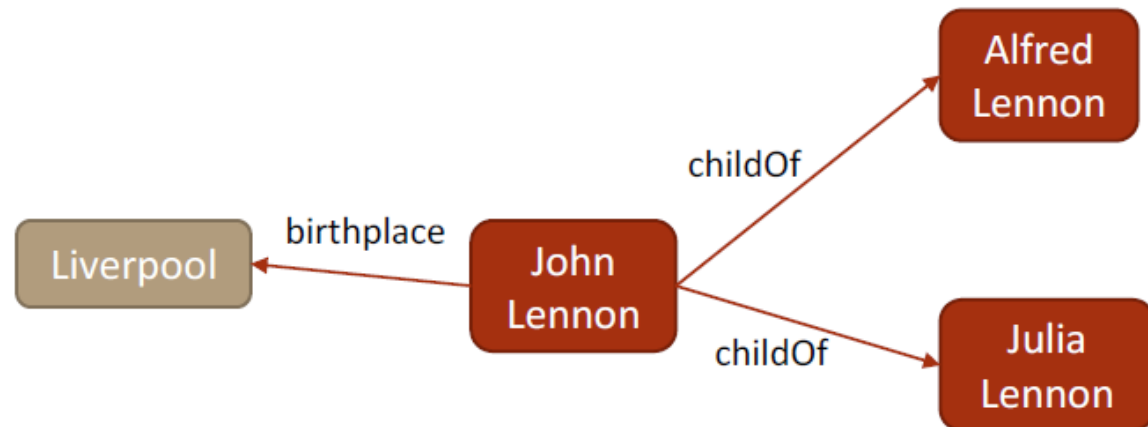
his father
he Alfred

Person Location Person Person
John was born in Liverpool, to Julia and Alfred Lennon.
NNP VBD VBD IN NNP TO NNP CC NNP NNP

Annotated text

**Information
Extraction**

Extraction graph



Breaking it down

Information
Extraction

Entity resolution,
Entity linking,
Relation extraction...



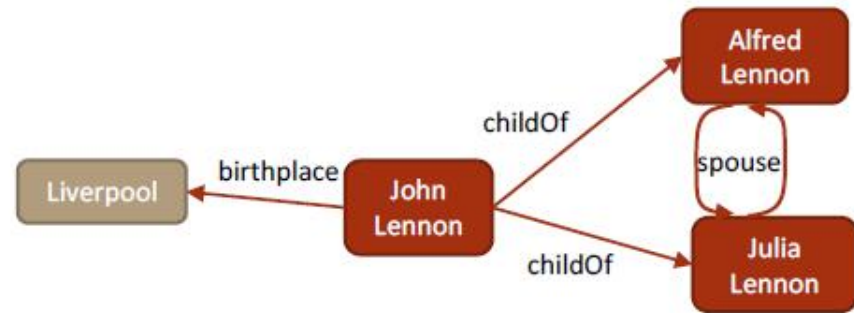
Document

Coreference Resolution...



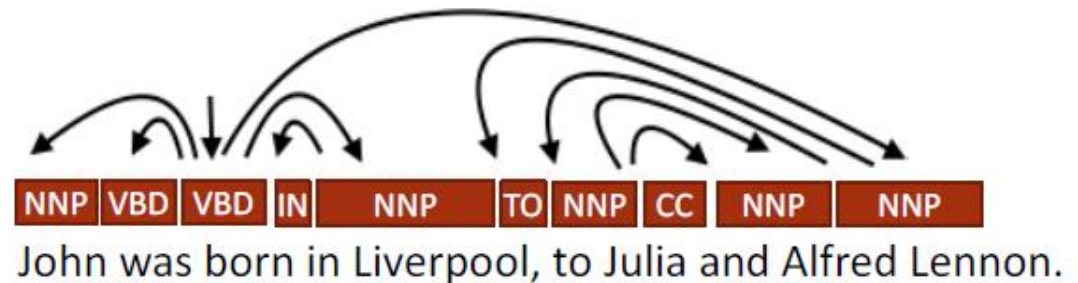
Sentence

Dependency Parsing,
Part of speech tagging,
Named entity recognition...

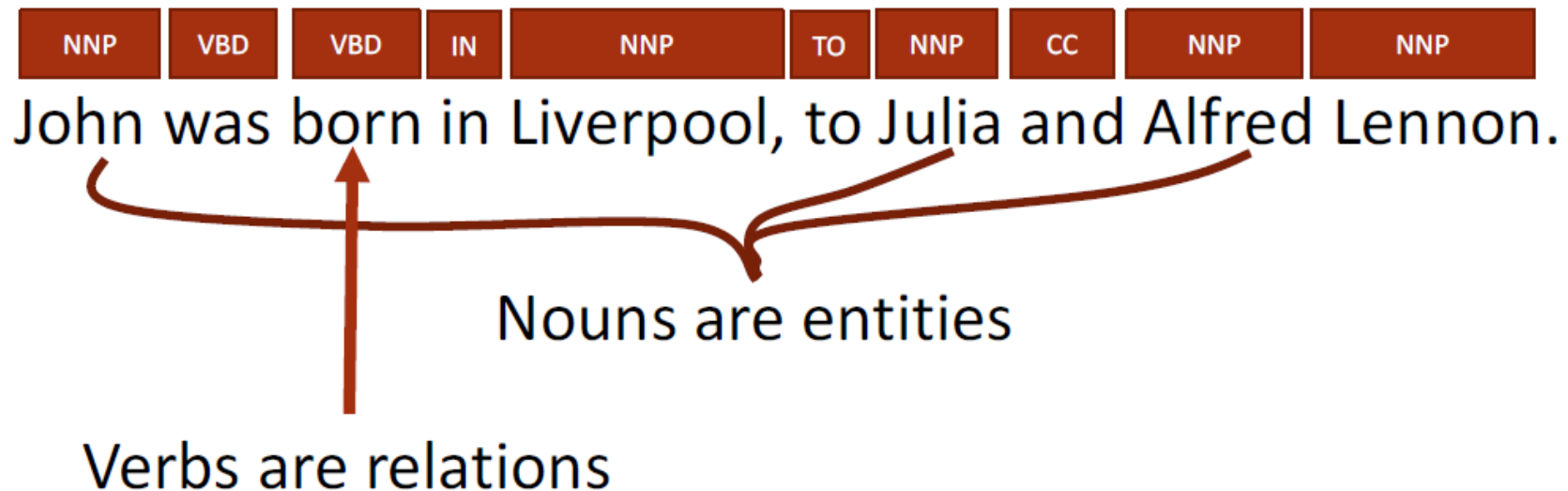


Lennon..
John Lennon... the Pool Mrs. Lennon..
 .. his mother .. his father
 he Alfred

Person Location Person Person
John was born in Liverpool, to Julia and Alfred Lennon.



Tagging the parts of speech (POS)



- POS resources:

- <https://parts-of-speech.info/>
- <http://nlp.stanford.edu:8080/parser/>
- Python packages: NLTK (<https://www.nltk.org/>), pyStatParser, TextBlob, Pattern, ...

Named entities recognition (NER)

Person Location Person Person
John was born in Liverpool, to Julia and Alfred Lennon.

- Machine learning for NLP:
 - Structured classification to add category label (e.g. Person, Location, Organization,...) for entity mentions and entity types
- NER resources:
 - <https://spacy.io/> (Python package: SpaCy)
 - <http://textanalysisonline.com/spacy-named-entity-recognition-ner>

Within-document co-reference

Same entity

He...
Lennon..
John Lennon...

Different
entities

the Pool

Same entity

Mrs. Lennon..
.. his mother ..

Same entity

Alfred
his father
he

John was born in Liverpool, to Julia and Alfred Lennon.

- Machine learning for NLP: Pairwise model for each noun/pronounce to consolidate information
 - Resources: <https://stanfordnlp.github.io/CoreNLP/>

Named entity resolution

...during the late 60's and early 70's, **Kevin Smith** worked with several local...

...the term hip-hop is attributed to **Lovebug Starski**. What does it actually mean...



Like Back in 2008, the Lions drafted **Kevin Smith**, even though Smith was badly...

... backfield in the wake of **Kevin Smith**'s knee injury, and the addition of Haynesworth...



The filmmaker **Kevin Smith** returns to the role of Silent Bob...

Nothing could be more irrelevant to **Kevin Smith**'s audacious ''Dogma'' than ticking off...



... The Physiological Basis of Politics," by **Kevin Smith**, Douglas Oxley, Matthew Hibbing...



Entity name: Two main problems

Entities with Same Name

Same type of entities share names

Kevin Smith, John Smith,
Springfield, ...

Things named after each other

Clinton, Washington, Paris,
Amazon, Princeton, Kingston, ...

Partial Reference

First names of people, Location
instead of team name, Nick names

Different Names for Entities

Nick Names

Bam Bam, Drumpf, ...

Typos/Misspellings

Baarak, Barak, Barrack, ...

Inconsistent References

MSFT, APPL, GOOG...

Entity resolution approach

Washington drops 10 points after game with UCLA Bruins.

Candidate Generation

Washington DC, George Washington, Washington state, Lake Washington, Washington Huskies, Denzel Washington, University of Washington, Washington High School, ...

Entity Types LOC/ORG

Washington DC, ~~George Washington~~, Washington state, Lake Washington, Washington Huskies, ~~Denzel Washington~~, University of Washington, Washington High School, ...

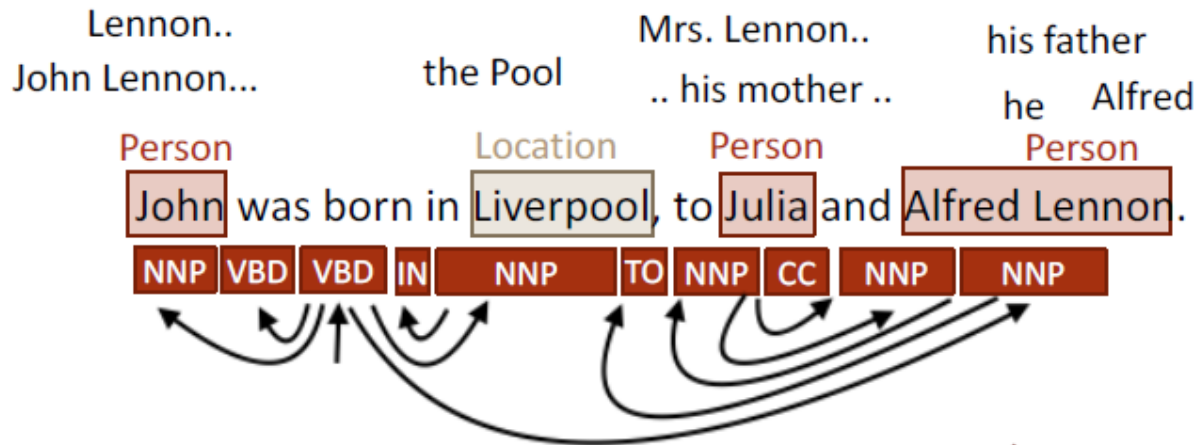
Coreference UWashington, Huskies

~~Washington DC~~, ~~George Washington~~, ~~Washington state~~, ~~Lake Washington~~, Washington Huskies, ~~Denzel Washington~~, University of Washington, ~~Washington High School~~, ...

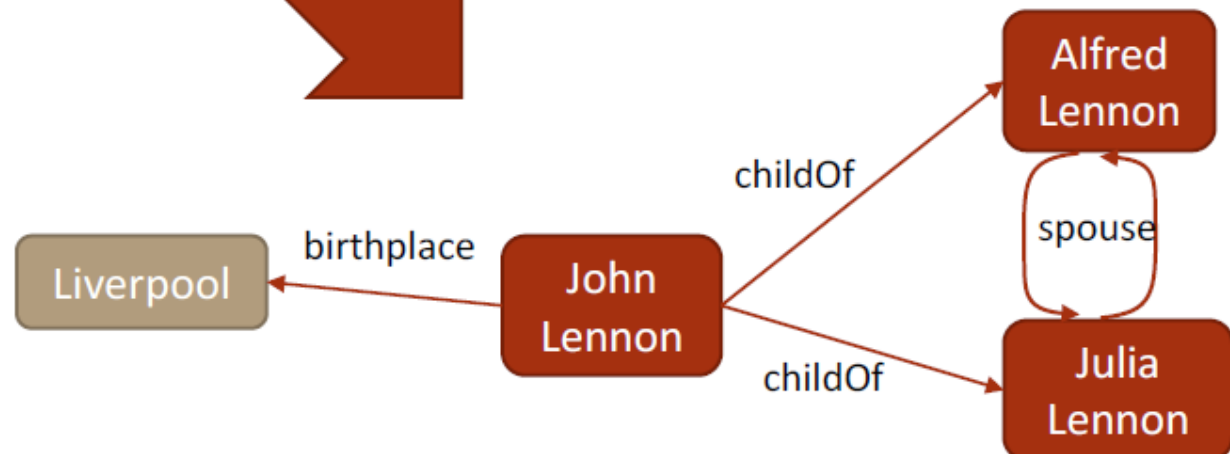
Coherence UCLA Bruins, USC Trojans

~~Washington DC~~, ~~George Washington~~, ~~Washington state~~, ~~Lake Washington~~, Washington Huskies, ~~Denzel Washington~~, ~~University of Washington~~, ~~Washington High School~~, ...

Knowledge extraction



Information Extraction



General framework and challenge

ID	Sentence
S1	Obama was born in Honolulu, Hawaii, USA as he has always said.
S2	President Clinton and Obama attended the funeral of former Israeli Prime Minister, and were scheduled to fly back to the US together.
S3	Barack Obama is the 44th and current President of the United States
S4	A dip of Barack Obama reading from his book " Dreams of My Father " has been shared out of context.
...	...

Automatically Labeled Training Data

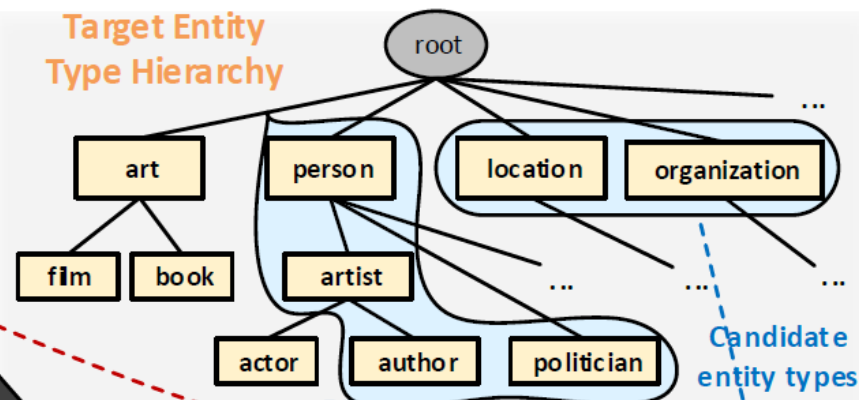
Relation Mention: ("Obama", "USA", S1)
 Types of Entity 1: {person, politician, artist, author}, Entity 2: {ORG, LOC}
 Relation Types: {president_of, bom_in, citizen_of, travel_to}

Relation Mention: ("Obama", "US", S2)
 Types of Entity 1: {person, politician, artist, author}, Entity 2: {ORG, LOC}
 Relation Types: {president_of, born_in, citizen_of, travel_to}

Relation Mention: ("Barack Obama", "United States", S3)
 Types of Entity 1: {person, politician, artist, author}, Entity 2: {ORG, LOC}
 Relation Types: {president_of, born_in, citizen_of, travel_to}

Relation Mention: ("Barack Obama", "Dreams of My Father", S4)
 Types of Entity 1: {person, politician, artist, author}, Entity 2: {book}
 Relation Types: {author_of}

Target Entity Type Hierarchy



Candidate relation types

Relation Type	Entity 1	Entity 2
president of	Barack Obama	United States
born in	Barack Obama	United States
citizen of	Barack Obama	United States
visit	Barack Obama	United States

KB Relations of Target Types

Information extraction

3 CONCRETE SUB-PROBLEMS

Defining domain

Learning extractors

Scoring the facts



3 LEVELS OF SUPERVISION

Supervised



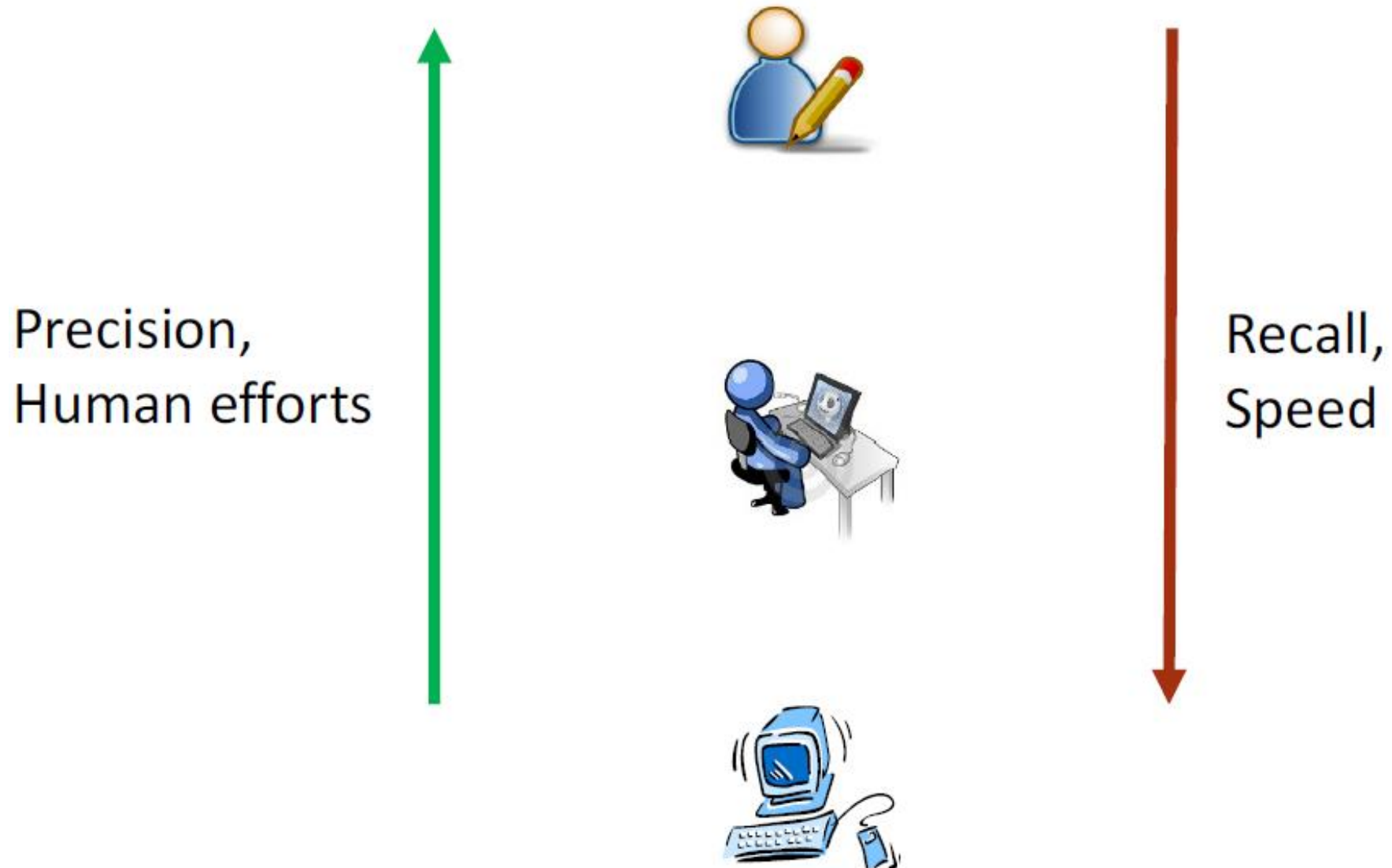
Semi-supervised



Unsupervised



Effect of supervision on extraction



Information extraction

3 CONCRETE SUB-PROBLEMS

Defining domain

Learning extractors

Scoring the facts



3 LEVELS OF SUPERVISION

Supervised



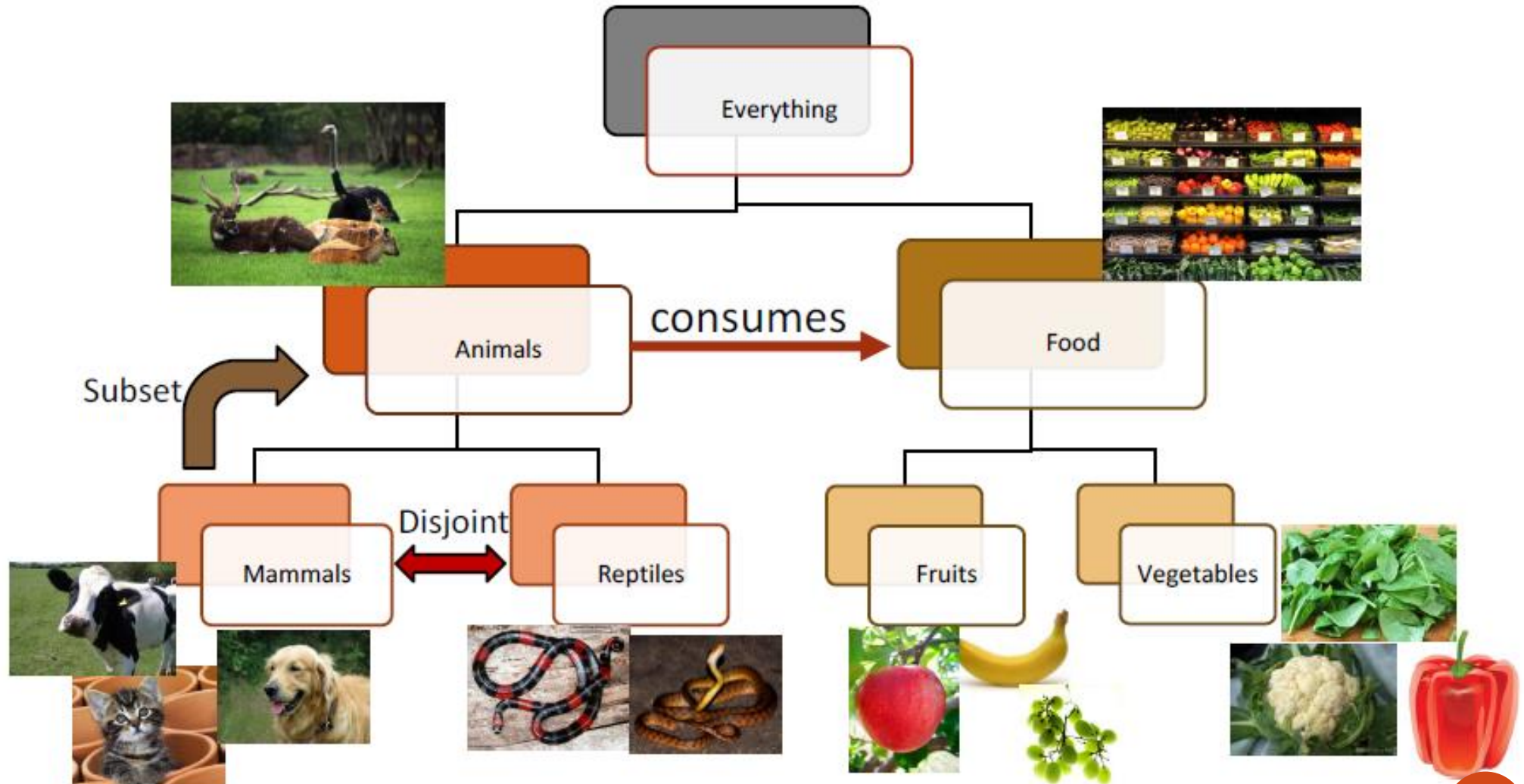
Semi-supervised



Unsupervised



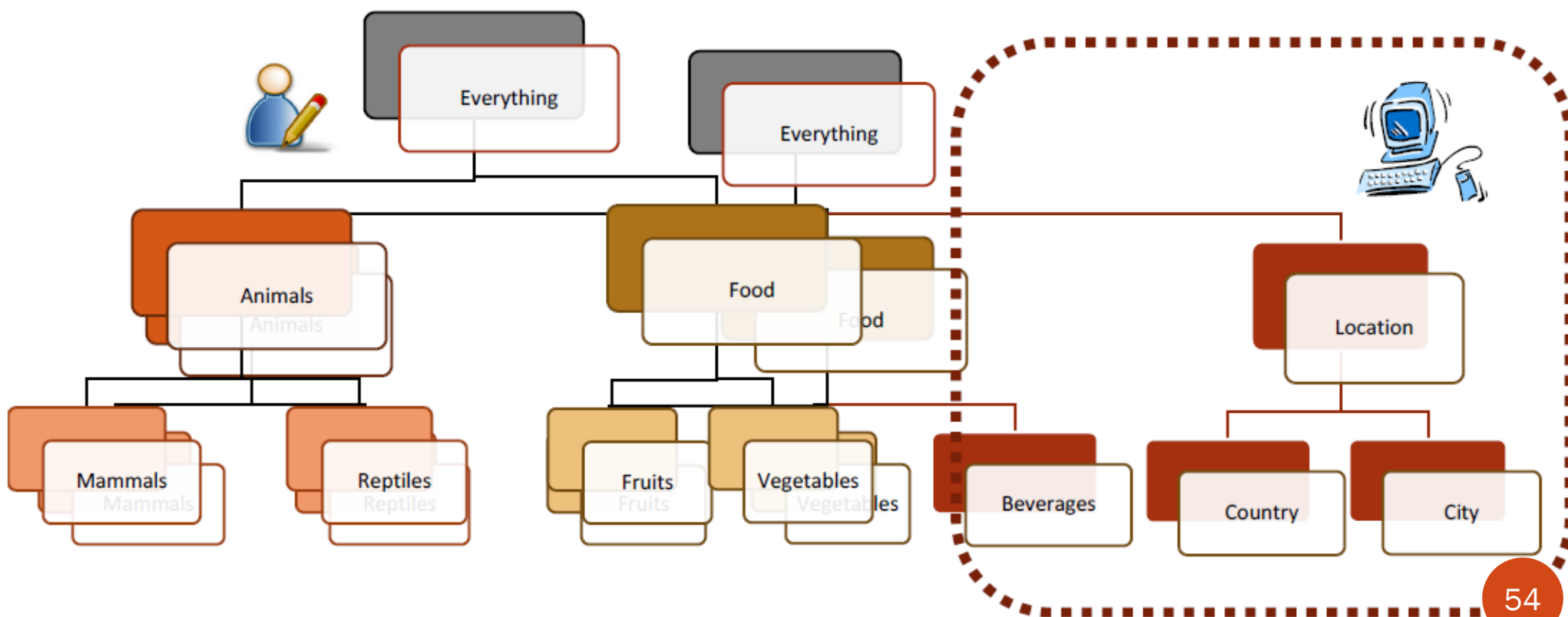
Defining domain: Manual



Defining domain: Semi-automatic



- Subset of types are manually defined
- SSL methods discover new types from unlabeled data



Define domain: Automatic



- Any noun phrase is a candidate entity
 - Dog, cat, cow, reptile, mammal, apple, greens, mixed greens, lettuce,...
- Any verb phrase is a candidate relation
 - Eats, feasts on, grazes, consumes,...

Information extraction

3 CONCRETE SUB-PROBLEMS

Defining domain

Learning extractors

Scoring candidate facts



3 LEVELS OF SUPERVISION

Supervised



Semi-supervised



Unsupervised



Learning extractors



- Supervised: high precision patterns
 - <PERSON> plays in <BAND>



- Semi-supervised: Bootstrapping to learn patterns
 - Create examples (John Lennon, Beatles), find patterns
 - Manually correct incorrect patterns



- Unsupervised: cluster phrases with constraints
 - Identify candidate verb phrases, find candidate arguments, cluster by NER types

Information extraction

3 CONCRETE SUB-PROBLEMS

Defining domain

Learning extractors



Scoring candidate facts

3 LEVELS OF SUPERVISION

Supervised



Semi-supervised



Unsupervised



Scoring the candidate facts



- Human defined scoring function or
Scoring function learnt using supervised ML with large amount of training data
{expensive, high precision}



- Small amount of training data is available
scoring refined over multiple iterations using both labeled and unlabeled data



- Completely automatic (Self-training)
 $\text{Confidence}(\text{extraction pattern}) \propto (\text{\#unique instances it could extract})$
 $\text{Score}(\text{candidate fact}) \propto (\text{\#distinct extraction patterns that support it})$
{cheap, leads to semantic drift}

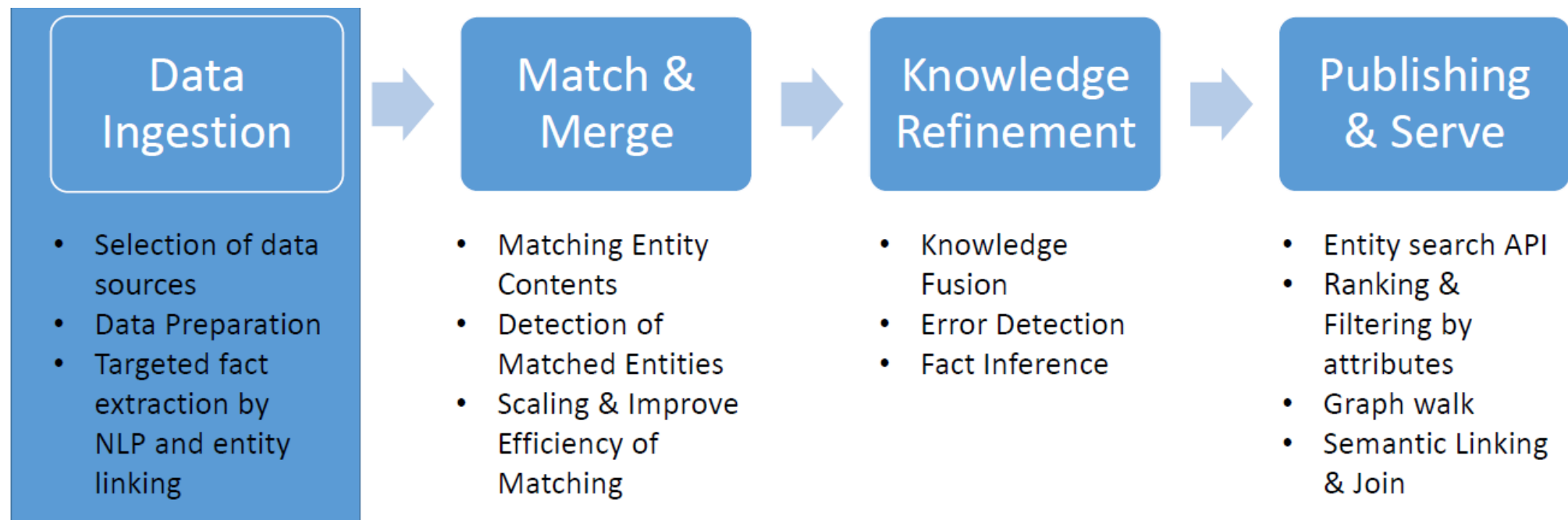
Information extraction in practice

	Defining domain	Learning extractors	Scoring candidate facts	Fusing extractors
ConceptNet				
NELL				Heuristic rules
Knowledge Vault				Classifier
OpenIE				

Knowledge extraction: Key points

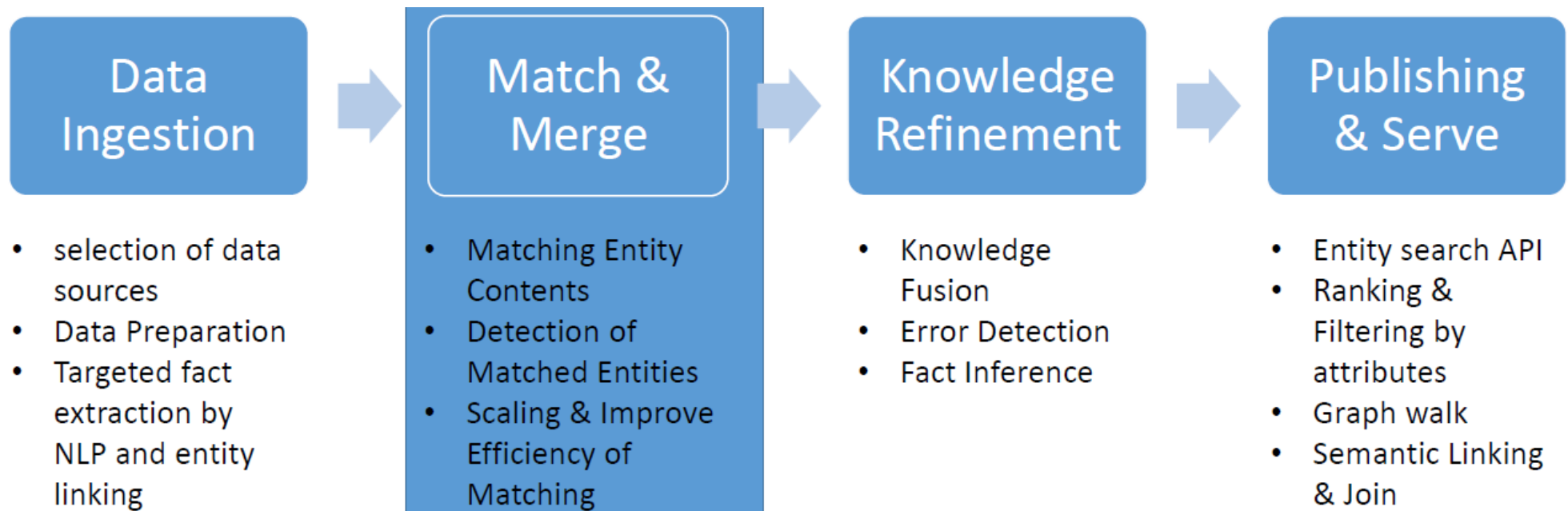
- Built on the foundation of NLP techniques
 - POS tagging, dependency parsing, named entity recognition, co-reference resolution...
 - Challenging research problems with very useful outputs
- Information extraction techniques using NLP and ML to define domains, extract entities and relations, score candidates
 - Trade-off between manual & automatic methods

Construct KG



Knowledge extraction

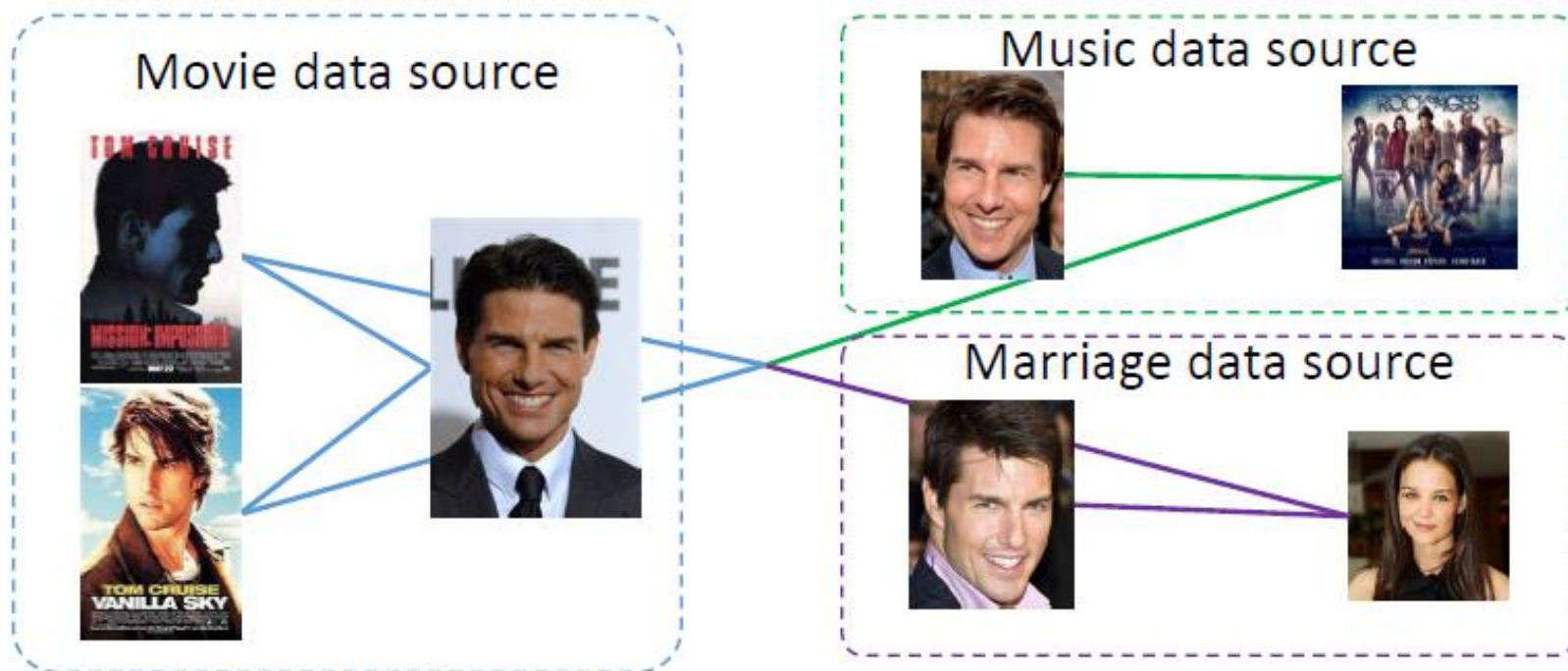
Construct KG



Big data challenges

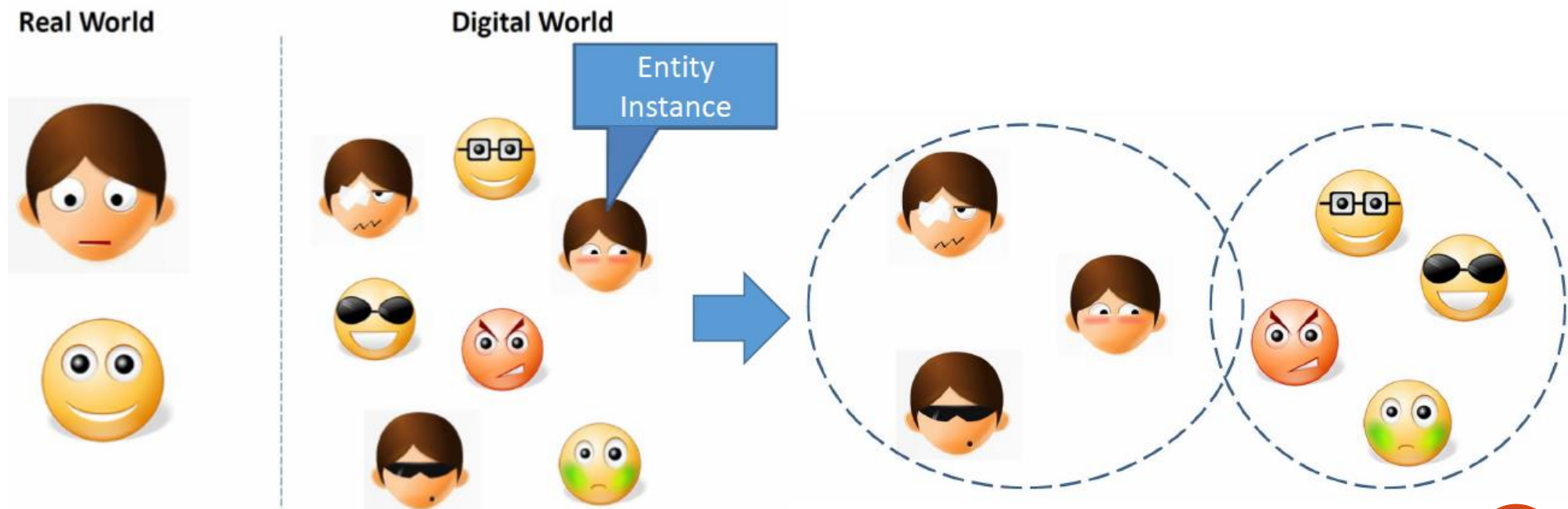
Entity matching (EM)

- Identify and discover instances referring to the same real-world entity to increase **data** quality
- **Synonyms**: Entity linking, entity resolution, reference reconciliation, deduplication, ...

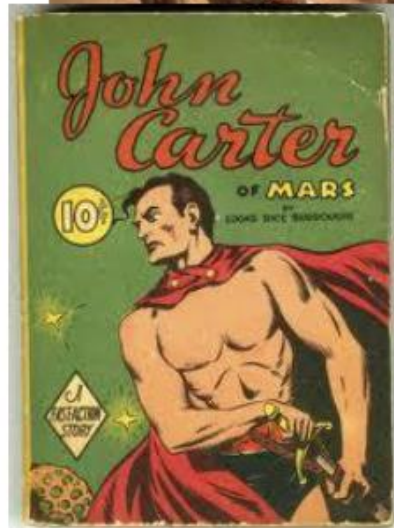


Data quality challenge

- Missing data
- Data error due to tech or human errors
- Abbreviations and truncation



Open Domain Entity Matching (Disambiguation Challenge)



John Carter   3rd
Quantitative Headhunter
Greater New York City Area | Fi



John Carter 3rd
**Head of Own Brand and Prod
Cash and Carry**
Munich Area, Germany | Consu



John Carter 3rd
Recruiter at STERIS Corporat
Cleveland/Akron Ohio Area | M

JOHN H. CARTER CO., INC.
"An Emerson Process Management Representative"

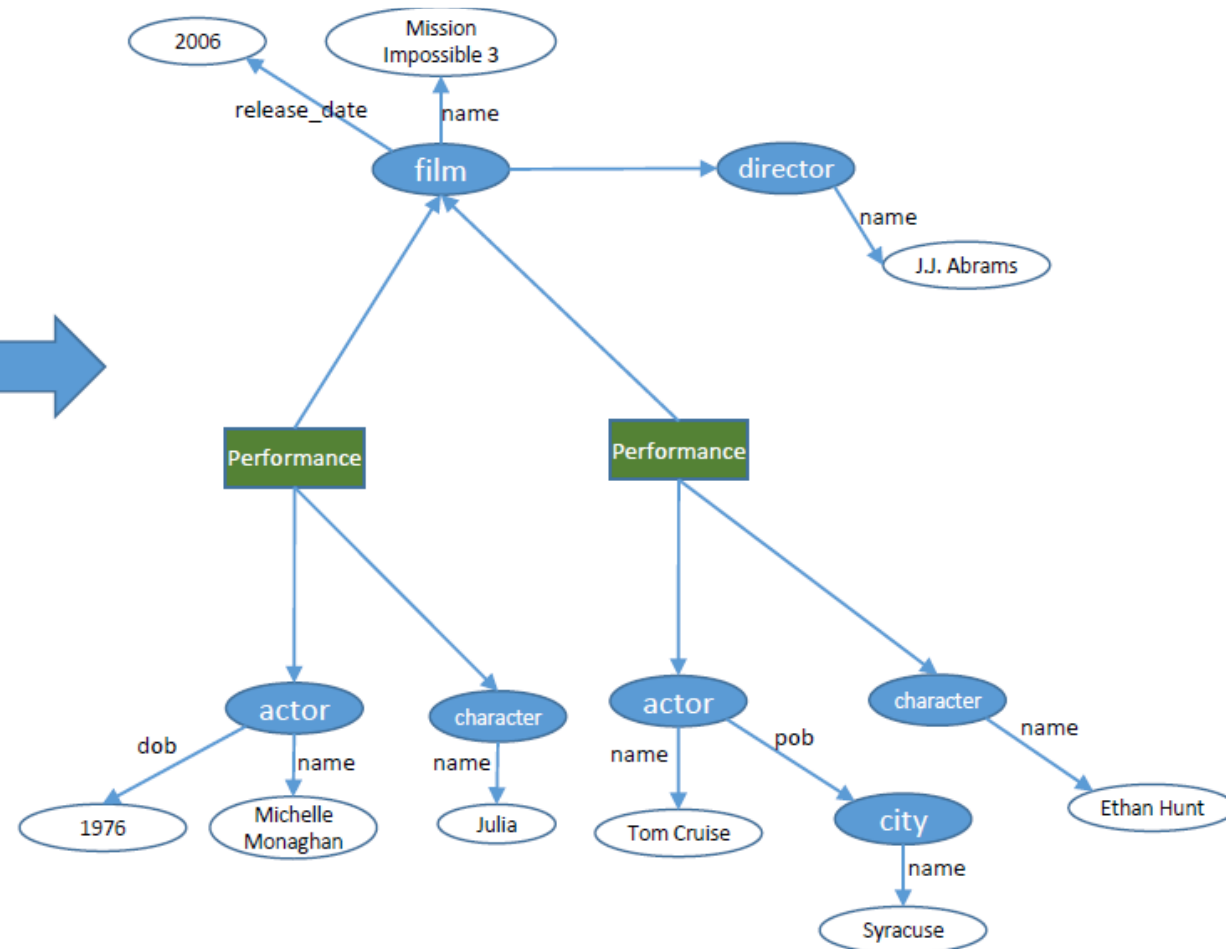
EM challenges in big data

- **Big data set:** Scalable, efficient, parallel techniques
- **Multi-domain:** Efficient techniques to manage executions within and across domains
- **Linked, connected and relational data:** Efficient techniques to leverage the diversity of connections and representation

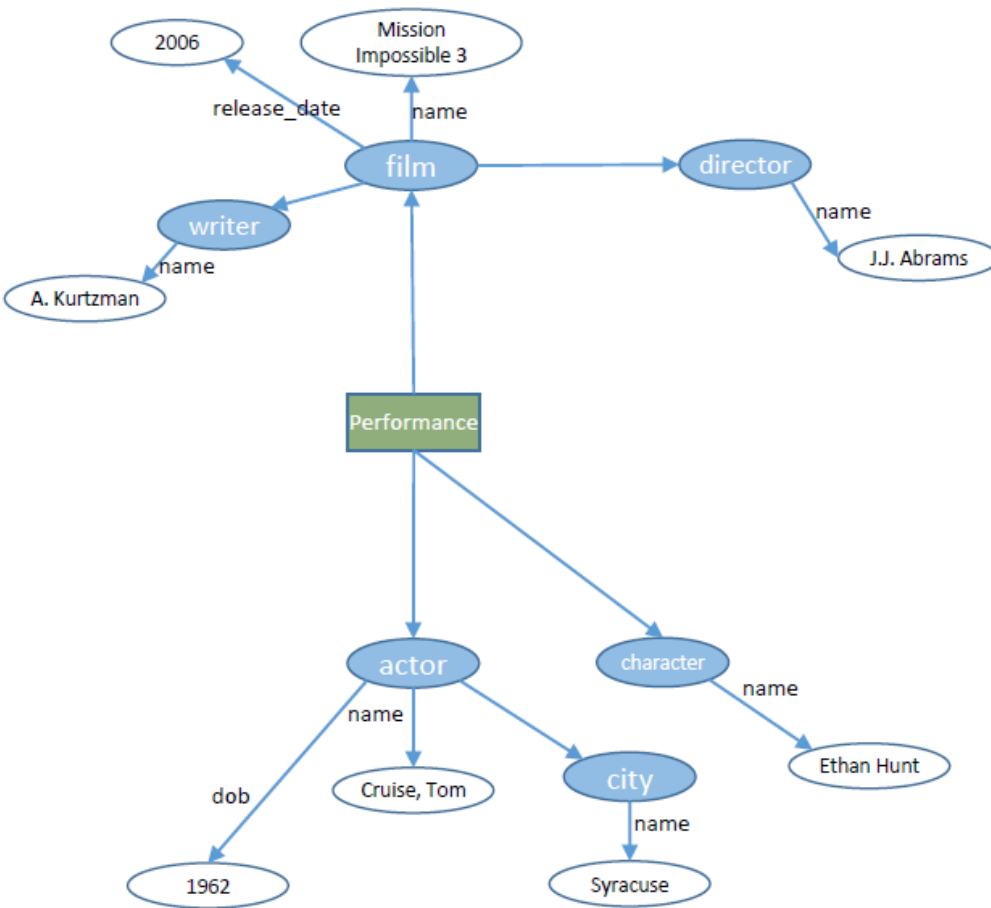
Techniques for EM

- Collective resolution in linked data
- Distance based models
- Probabilistic matching models (not covered)
 - Supervised and semi-supervised learning
 - Active learning
- Declarative matching rules and constraints (not covered)
 - Disjunction of conjunction rules
 - Constraint-based clustering

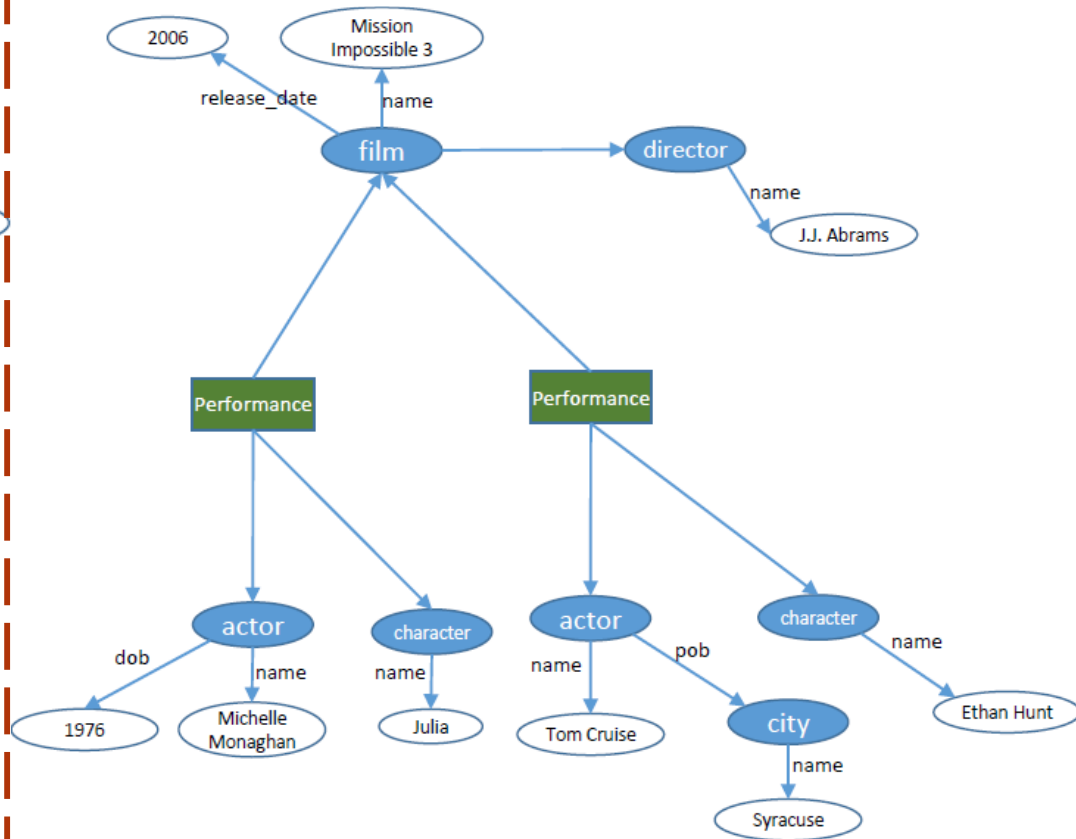
Collective resolution in linked data



Graph model and conflation

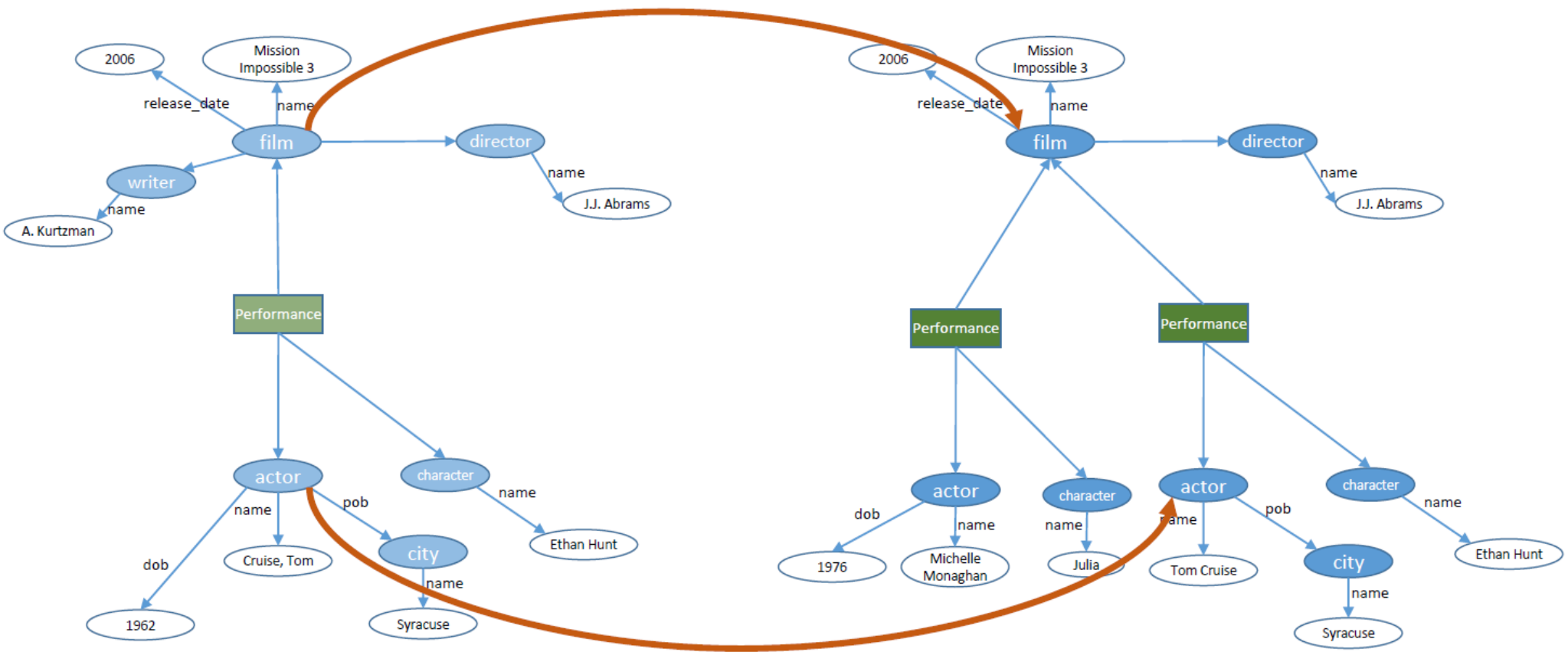


Database 1

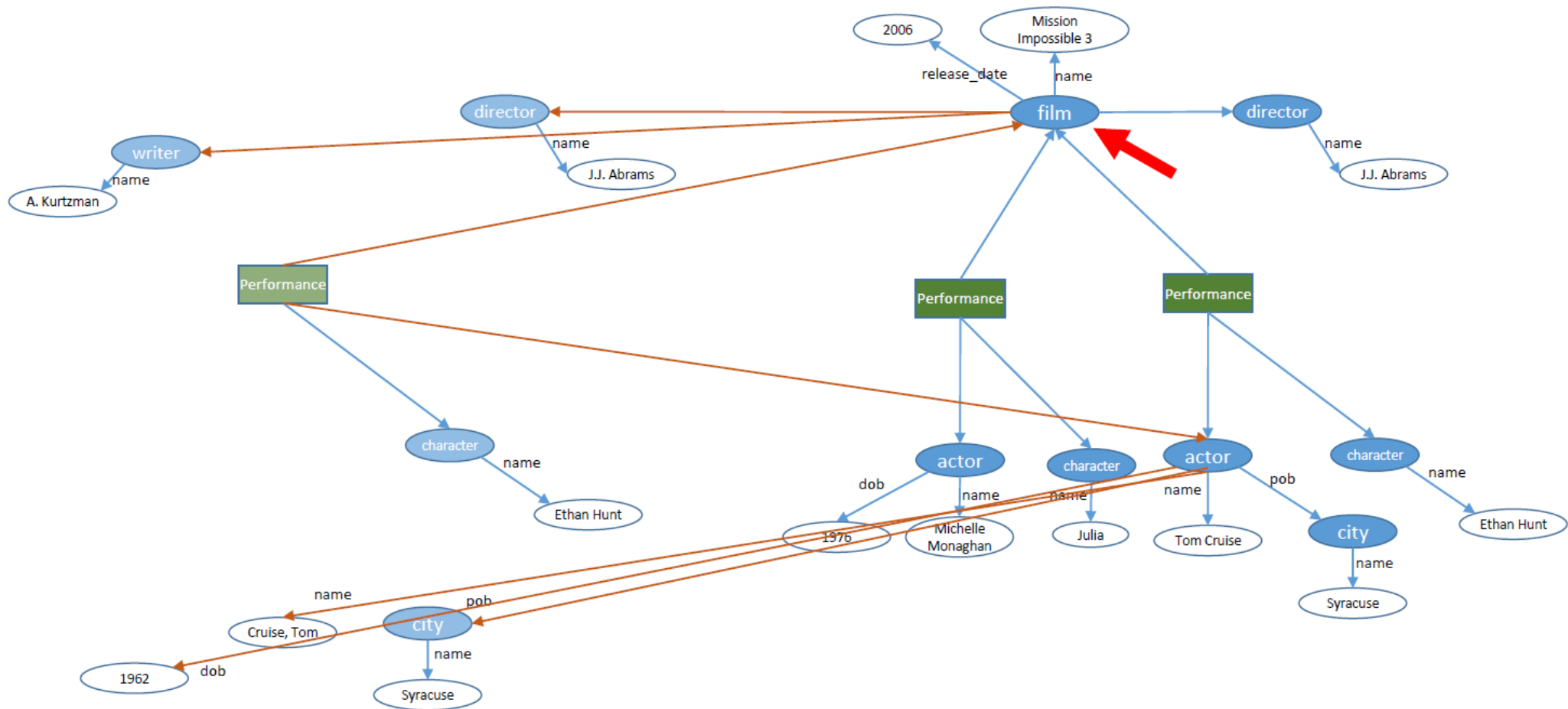


Database 2

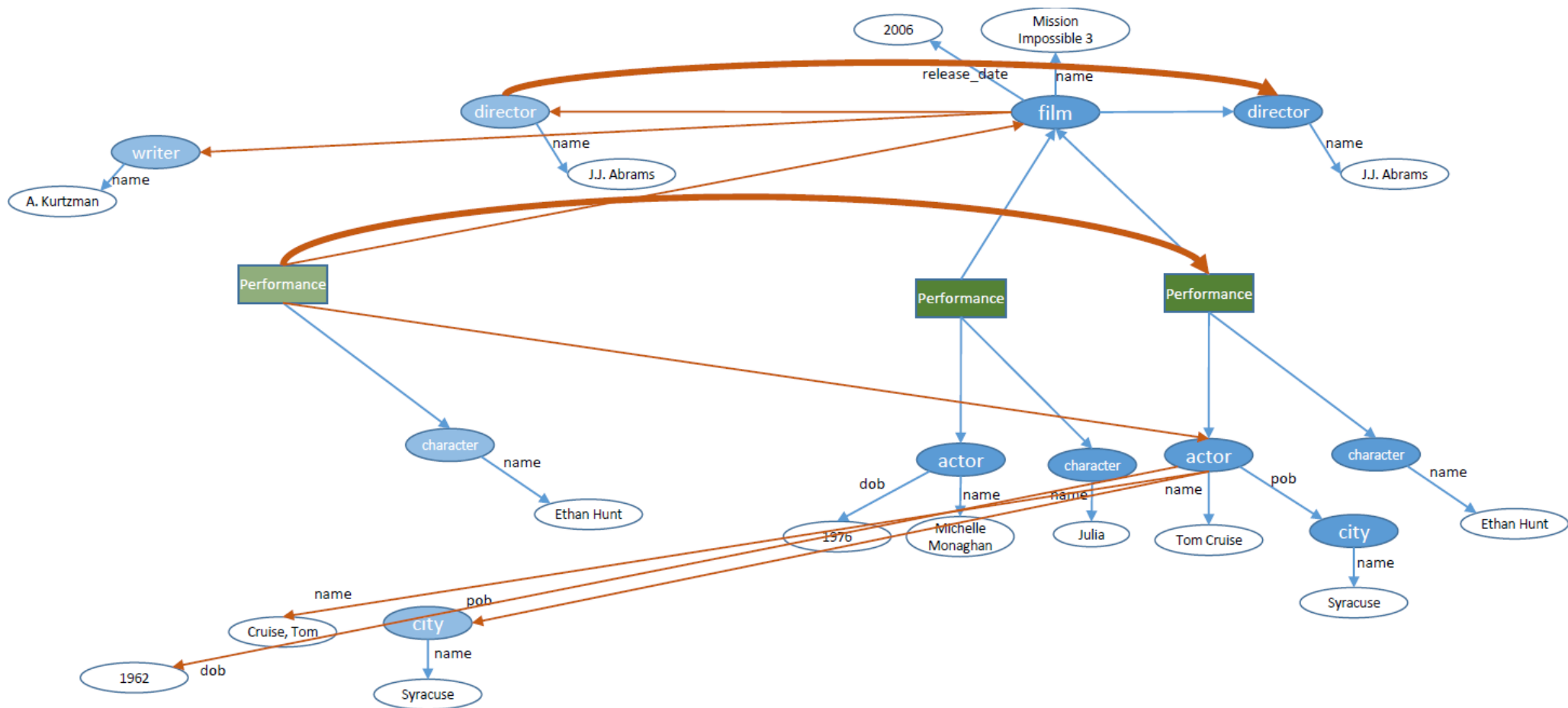
Graph model and conflation



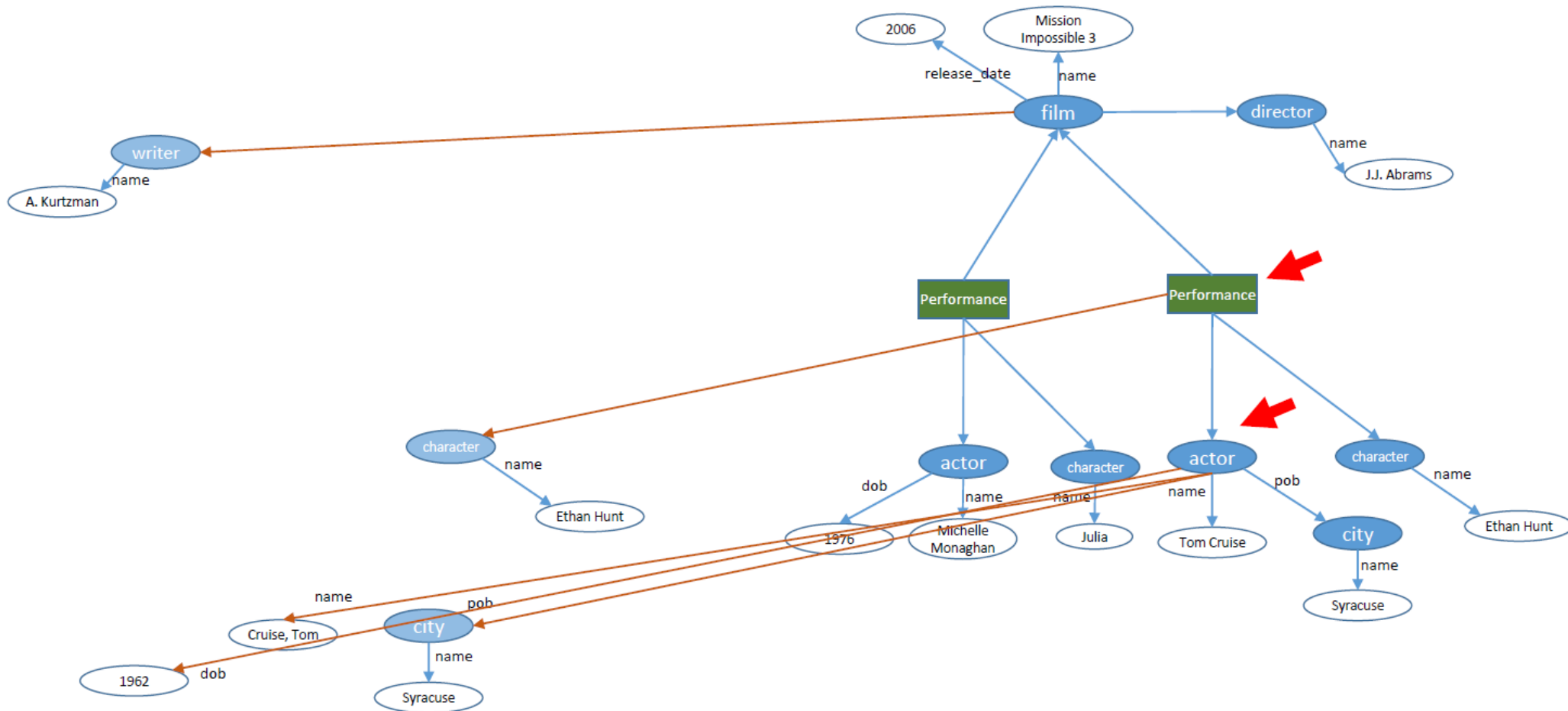
Graph model and conflation



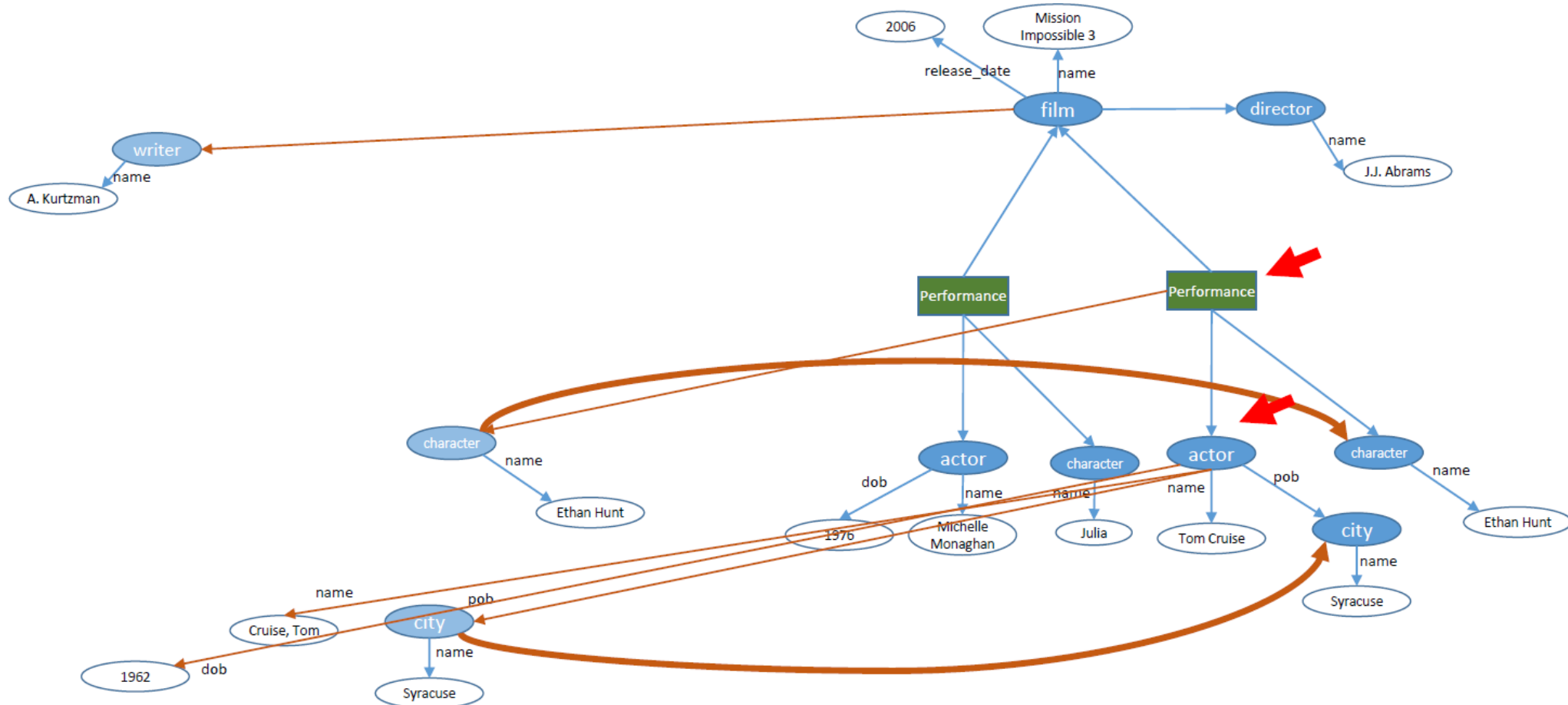
Graph model and conflation



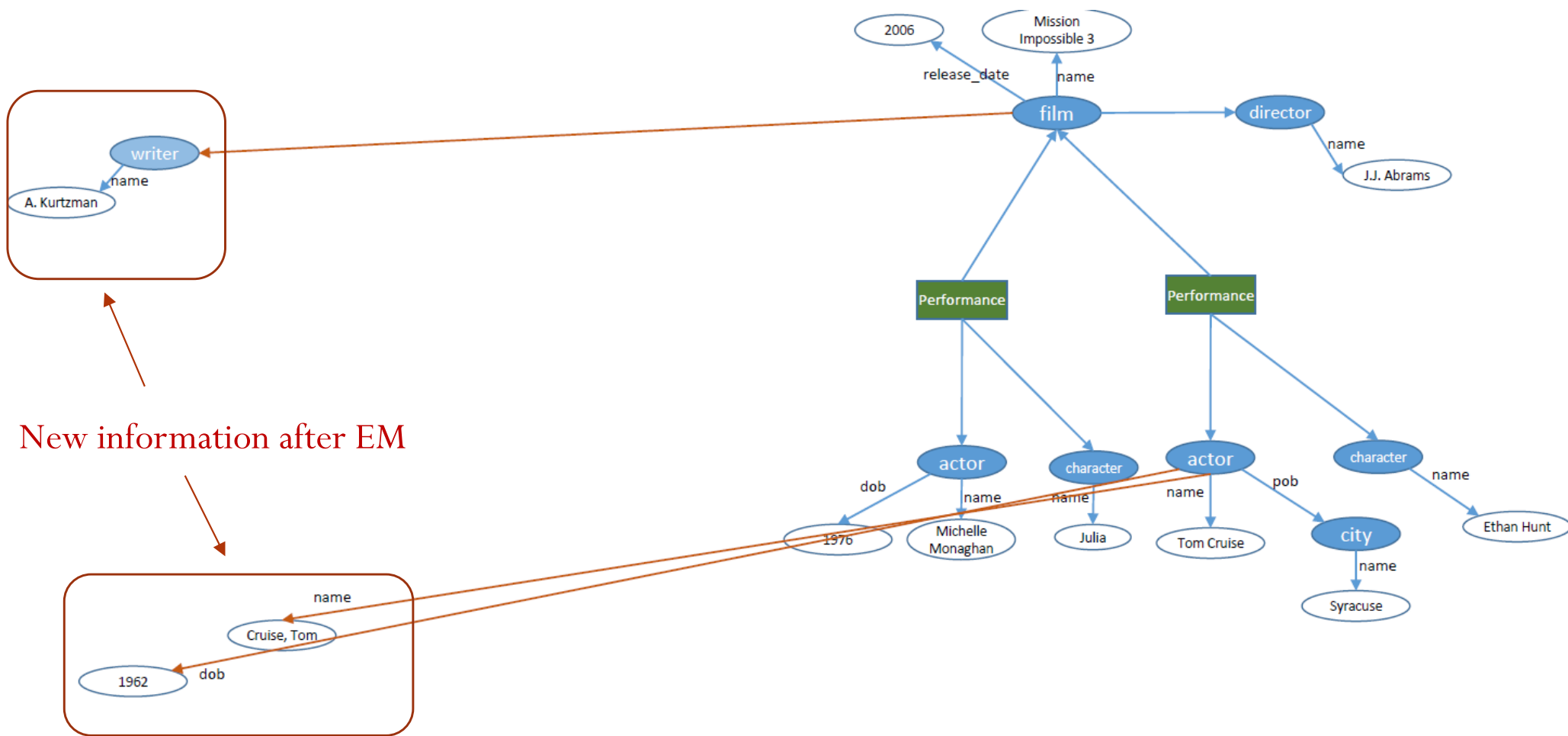
Graph model and conflation



Graph model and conflation



Graph model and conflation



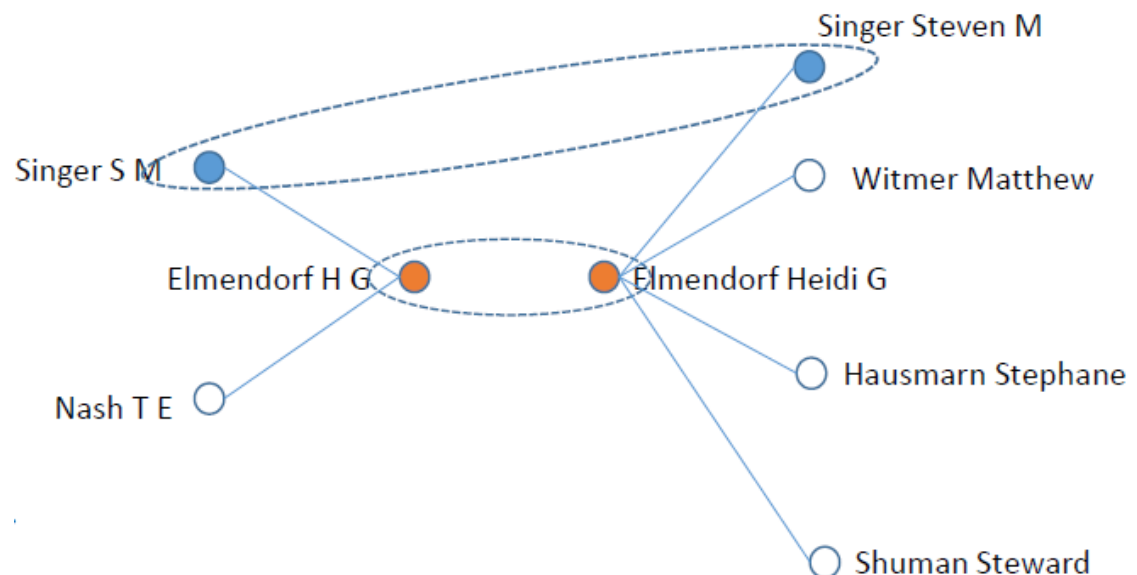
Collective resolution in linked data

- Entity similarity based on connection
- Measures:
 - **SimRank**: Two nodes are similar if they are referenced by similar nodes

$$sim(a, b) = \frac{c}{|I(a)||I(b)|} \sum_{i=1}^{|I(a)|} \sum_{j=1}^{|I(b)|} sim(I_i(a), I_j(b))$$

where **I(a)** is the neighbors of the node **a**

- Adar measure
- Katz score
- ...



Distance-based EM models

- Threshold method:

- Compute similarity vector using some similarity measure
- Classify the vectors as Match if the aggregate similarity is larger than a **threshold t** ; otherwise, UnMatch

- Example:

- Similarity vector $\langle 0.8, 0.7, 1.0 \rangle$
- Verify: $\text{sim}(\text{Name}_1, \text{Name}_2) + \text{sim}(\text{Prof}_1, \text{Prof}_2) + \text{sim}(\text{Add}_1, \text{Add}_2) > t$

Name	<u>John</u> Smith
Profession	Software Eng.
Address	Seattle

Name	<u>Johan</u> Smith
Profession	Software <u>Dev</u> Eng.
Address	Seattle

Similarity



Vector

Name	0.8
Profession	0.7
Address	1.0

Distance-based EM models

- Scalability problem:
 - Matching two data sources each with **1M** entities
 - **1M x 1M** with an entity pair comparison time of **5 us**
 - **160 years**
 - **300K** machines to finish in **5 hrs**
- Solution:
 - Indexing
 - Locality-sensitive hashing (CS 753)
 - MapReduce-based solution (MapReduce lecture)

Distance-based EM models

- Inverted index:

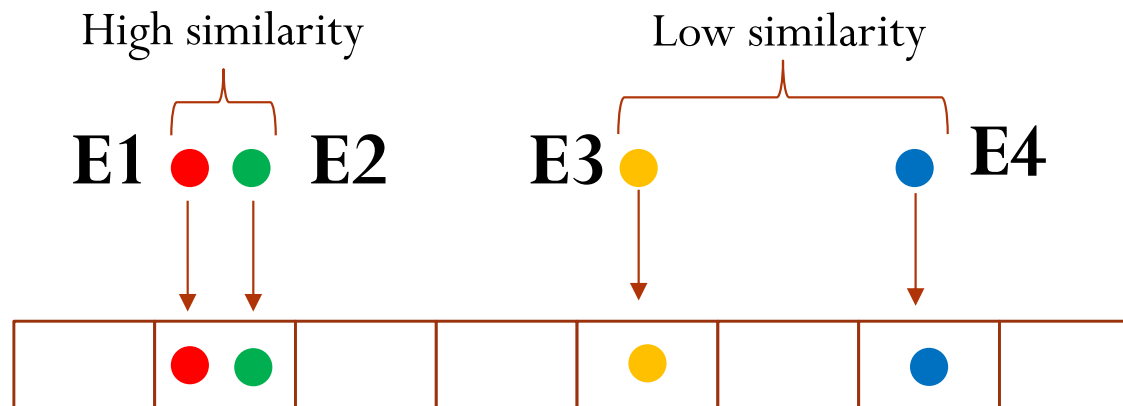
Entity	Features		Key	Post list
E1	F1, F2		F1	E1, E2
E2	F1, F3, F4	→	F2	E1
E3	F3		F3	E2, E3
E4	F4		F4	E2, E4

- Why efficient?

- Candidates (E1, E2), (E2, E3), (E2, E4) compared to all possible 6 pairs (E1, E2), (E1, E3), ..., (E3, E4)
- Can extend to idf weight for text/document

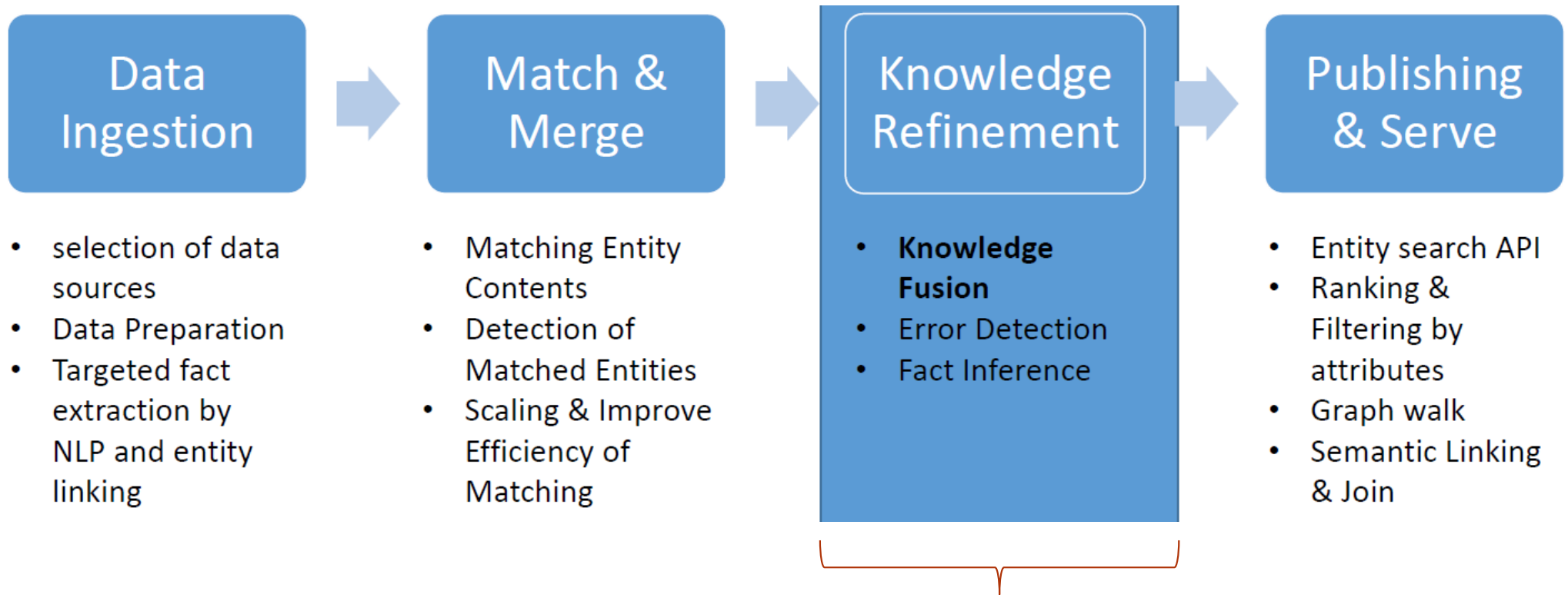
Distance-based EM models

- Locality-sensitive hashing:



- Why efficient?
 - Only check candidate pairs (E1, E2) in the bucket compared to all possible 6 pairs (E1, E2), (E1, E3), ..., (E3, E4)
 - Can extend to entities (images, text) with metric distance

Construct KG



Post-processing after merging entities

Knowledge fusion

- After merging entity nodes, there are conflicting facts and connections
- **Solutions:** Majority voting

	Name	Birth date	Birthplace	Age	Children
S1	Barack Obama	August 4, 1961	Hawaii, US	56	Malia, Sashia
S2	Barack Hussein Obama	08/04/1961	Honolulu	56	Malia
S3	Barack H. Obama	07/04/1961	Kenya	56	Ivanka



Majority voting and correctly identify similar values. Accuracy < 1.0

Knowledge fusion

- After merging entity nodes, there are conflicting facts and connections
- **Solutions:** Identify authoritative source

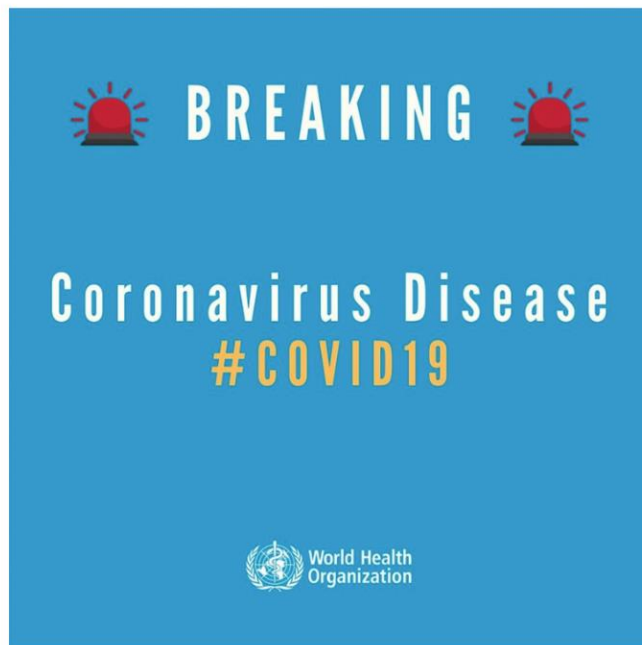
	Name	Birth date	Birthplace	Age	Children
S1	Barack Obama	August 4, 1961	Hawaii, US	56	Malia, Sashia
S2	Barack Hussein Obama	08/04/1961	Honolulu	56	Malia
S3	Barack H. Obama	07/04/1961	Kenya	56	Ivanka



Accurate source, Accuracy = 1.0

Knowledge fusion

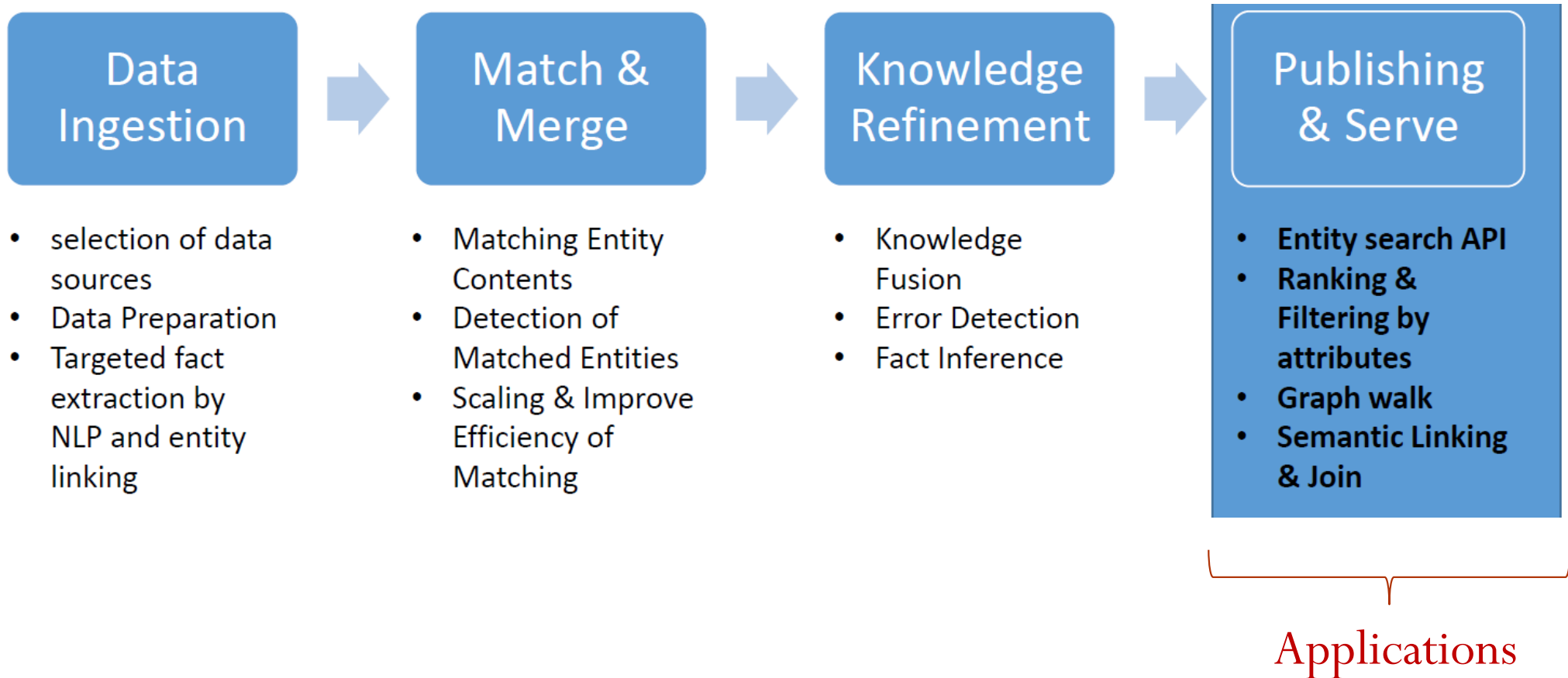
- After merging entity nodes, there are conflicting facts and connections
- Solutions:
 - Fact checking tutorial (<https://shiralkarprashant.github.io/fact-checking-tutorial-KDD2018/>): not only for KGs but also for **fake news**



Error detection and fact inference

- Error detection using data quality rules:
 - Functional dependency (e.g. Zip $\rightarrow$ City)
 - Inconsistency (e.g. data_of_birth > date_of_death)
 - Outlier detection from graph
- Fact inference:
 - Further enrichment or data completion by internal information (e.g. dominant type and label) or external information (e.g. feedback)

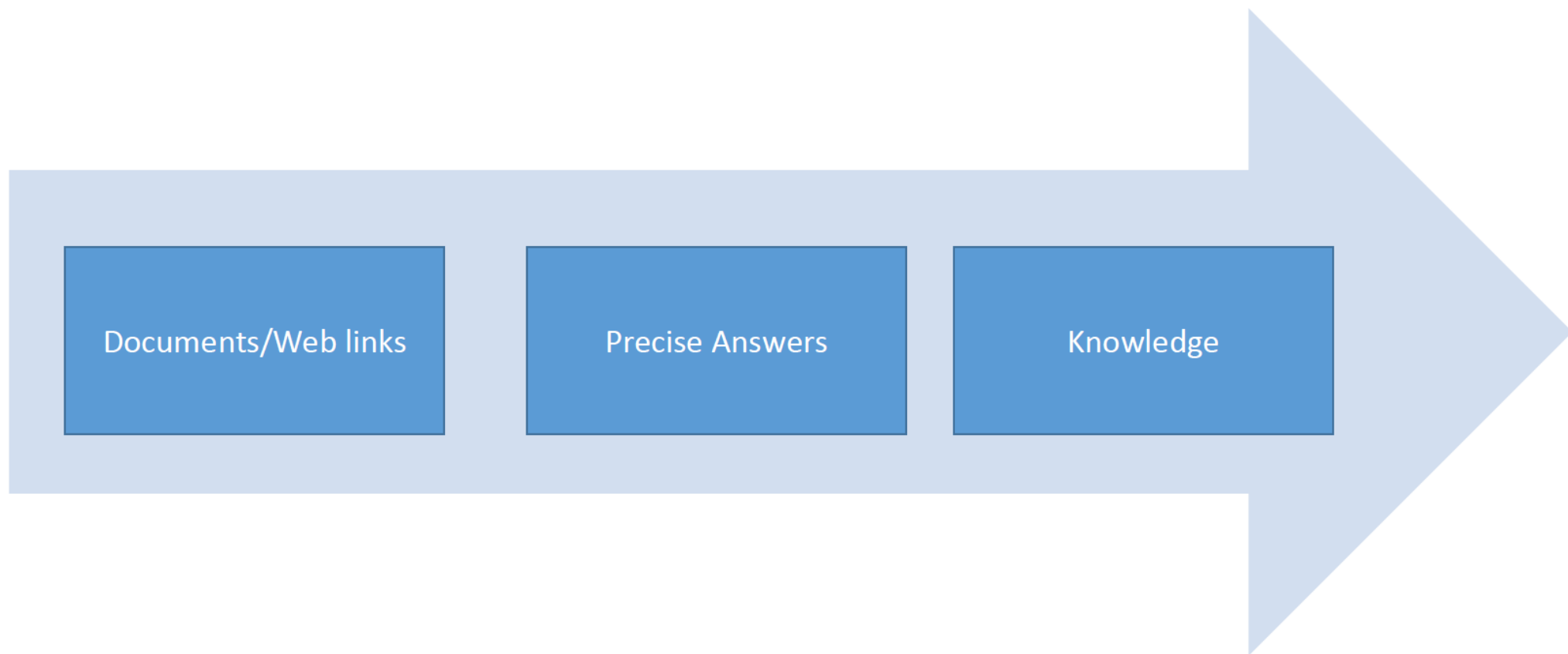
Construct KG



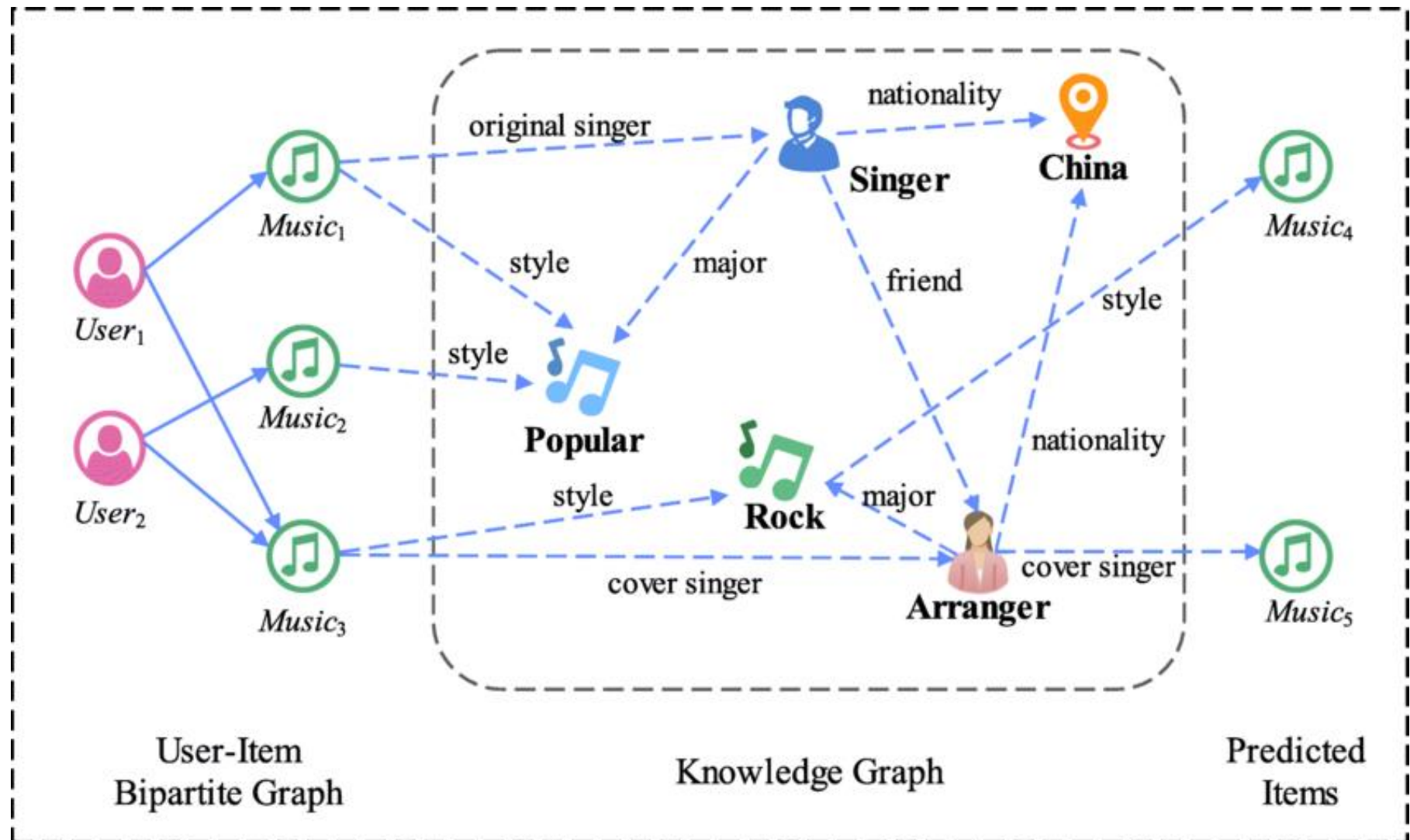
Outline

- Introduction of knowledge graph
- How to construct knowledge graph
- Challenges and success stories

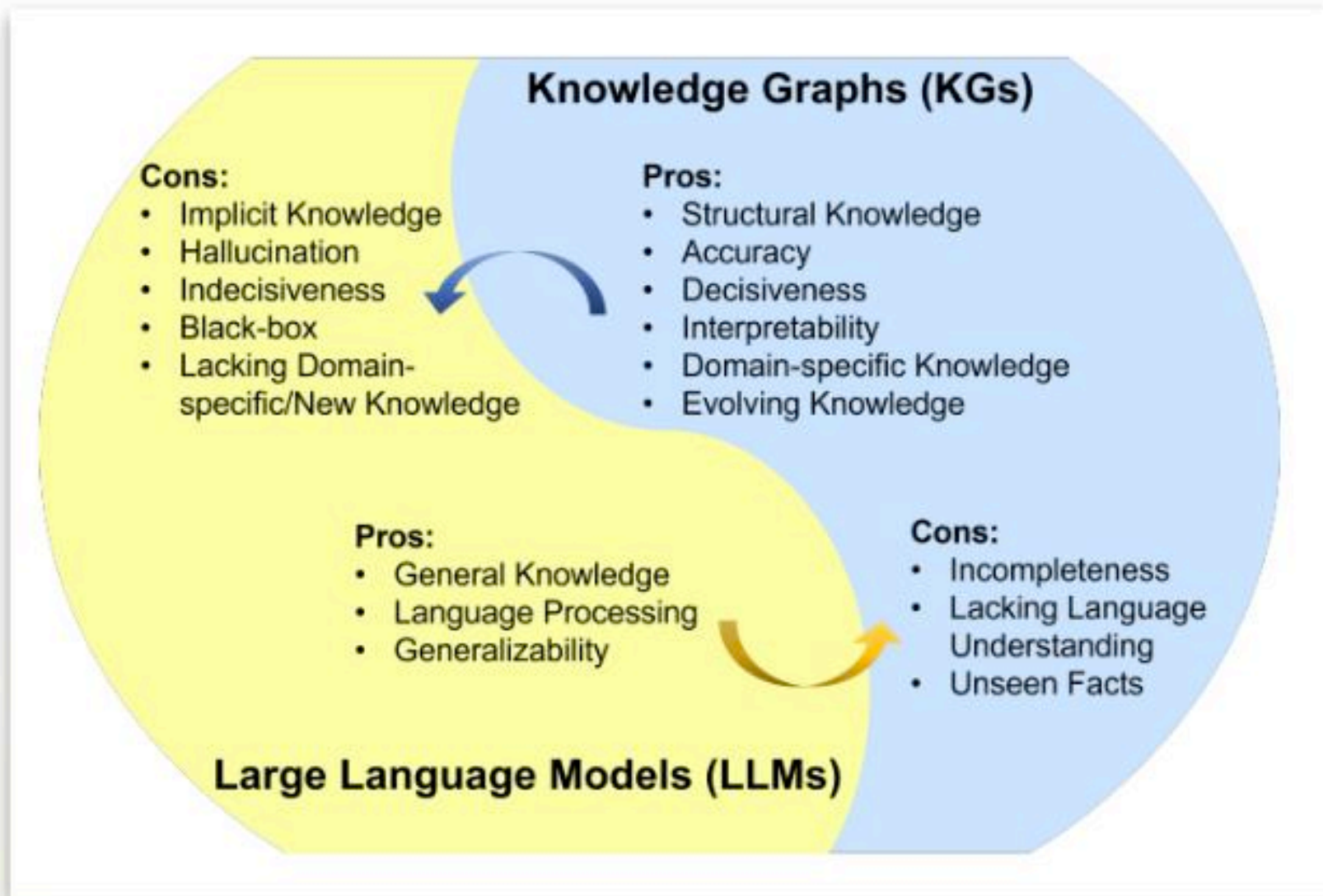
Why KGs serve?



KGs used in recommender systems



Fusion of KGs and LLMs



Challenges of scaled KGs

Coverage

Have we got the information we need?

Freshness

Is information up to date?

Correctness

Is our information accurate?

Three forces in constant conflict:

Increased freshness and coverage



Harder to ensure correctness

Increased correctness



Harder to ensure freshness and coverage

Correctness is *always* hard – what is true and correct?
Particularly critical in today's world



Will Smith
American Actor

Willard Carroll Smith Jr. is an American actor and songwriter. In April 2007, he was named "Actor in Hollywood". Smith has been nominated for two Academy Awards and two Golden Globes.

Wikipedia IMDb Facebook

Born: Sep 25, 1968 (age 49) · P
Height: 6' 2" (1.88 m)
Spouse: Jada Pinkett Smith (m.)
Children: Jaden Smith (Son) · Will
Upcoming movies: Bad Boys III
4 · Gemini Man



Will Smith
Baseball

William Michael Smith
San Francisco Giants
for the Kansas City

Wikipedia Twitter

Born: Jul 10, 1968
Height: 6' 5" (1.96 m)
Team: San Francisco
Salary: \$2.00 million
Education: Gulf Coast Community College

Will Smith
Defensive End

William Raymond Smith III was an American football defensive end in the National Football League. He played college football for Ohio State and was drafted by the New Orleans Saints in the first round of the 2004 NFL draft, where he played for the entire...

Wikipedia

Lived: Jul 04, 1981
Height: 6' 3" (1.91 m)
Salary: \$3 million
40-yard dash time
Spouse: Raquel Jones
Parents: William S.

Will Smith
Comedian

Follow my progress on Twitter and Instagram: Twitter: @Will_Smith Instagram: @Will_Smith

Facebook: 1.9k followers
Twitter: 3.4k followers
Data from: Facebook
Suggest an edit

Will Smith
Comedian

William James Smith is an English stand-up comedian, screenwriter, novelist, actor and producer. As co-writer and co-producer of the critically acclaimed HBO sitcom Veep, he was among the recipients of two Emmys and two Writers Guild...

Wikipedia

Born: Jun 08, 1971 (age 48) · Winchester, England
Siblings: Cilly Smith (brother)
Awards: Writers Guild of America Award for Television: Comedy Series (2014)
Nominations: Primetime Emmy Award for Outstanding Comedy Series (2014) · Writers Guild of America Award for Television: New Series

Movies and TV shows
See all (10+)

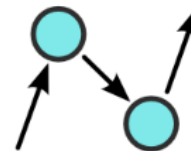
Will Smith: Single entity, 108K facts assembled from 41 web sites. There are 200 Will Smiths on Wikipedia alone.

Success stories

openie.allenai.org



Open Information Extraction



ConceptNet

An open, multilingual knowledge graph

NELL Knowledge Base Browser

CMU Read the Web Project



Open Information Extraction



Argument 1: Beatles

Relation:

Argument 2:

All ▾

Search

302 answers from 2640 sentences (cached)

all

location (27)

film location (25)

statistical region (22)

name source (22)

governmental jurisdiction (19)

misc.

more types ▾

were bigger than **Jesus** (104)

came to America (95)

appeared on **The Ed Sullivan Show** (81)

Here Comes the Sun (64)

broke up in 1970 (56)

are a great band (50)

came to America (45)

perform on **The Ed Sullivan Show** (39)

were **Musical ensemble** (36)

are now on **iTunes** (29)

released their first album (28)

came on the scene (27)

citytown (19)

tv actor (19)

employer (18)

person (18)

hall of fame inductee (16)

organization scope (15)

deceased person (15)

fictional setting (14)

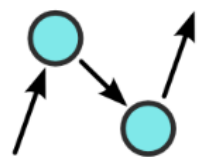
artist (14)

actor (13)

award winner (12)

person or entity appearing in film (12)

place of interment (11)



ConceptNet

An open, multilingual knowledge graph

en beatles

An English term in ConceptNet 5.7

Sources: Open Mind Common Sense contributors, DBpedia 2015, JMDict 1.07, OpenCyc 2012, English Wiktionary, French Wiktionary, and Open Multilingual WordNet

[View this term in the API](#)

[Documentation](#)

[FAQ](#)

[Chat](#)

Synonyms

- [fi](#) Beatles (n, group) →
- [fr](#) Beatles (n, group) →
- [ja](#) ビートルズ (n) →
- [ja](#) ビートルズ (n, group) →
- [th](#) วงเดอะ บีเทิลส์ (n, group) →
- [ar](#) بيتلز →
- [ar](#) بيتلز →
- [az](#) bitlz →
- [be](#) бітлз →
- [bg](#) бийтълс →
- [bn](#) দ্য বিটল্‌স →
- [bn](#) বিটল্‌স →
- [el](#) μπητλσ →

Related terms

- [en](#) sixties (n, time) →
- [en](#) beatle (n) →
- [en](#) beatlehead (n) →
- [en](#) beatlemania (n) →
- [en](#) british →
- [en](#) operate →
- [en](#) quartet →
- [en](#) rock music →
- [fr](#) beatles →
- [en](#) ex beatle (n) →
- [en](#) lennon (n) →
- [en](#) mccartney (n) →
- [es](#) beatles (n) →

Links to other sites

- [dbpedia.org](#) The_Beatles →
- [umbel.org](#) Beatle →
- [wikidata.dbpedia.org](#) Q1299 →
- [sw.opencyc.org](#) Beatle →
- [wordnet-rdf.princeton.edu](#) 400520405-N →
- [wordnet-rdf.princeton.edu](#) 108386847-n →
- [dbpedia.org](#) The_Beatles_(No._1) →
- [wikidata.dbpedia.org](#) Q738260 →
- [dbpedia.org](#)
The_Beatles_(The_Original_Studio_Recordings) →
- [wikidata.dbpedia.org](#) Q603122 →
- [en.wiktionary.org](#) Beatles →
- [fr.wiktionary.org](#) Beatles →

Derived terms

- [en](#) beatle →
- [en](#) beatledom →
- [en](#) beatlemania →
- [en](#) Beatlesque →
- [en](#) ex beatle →
- [en](#) fourth beatle →

YAGO: A High-Quality Knowledge Base

YAGO is a large knowledge base with general knowledge about people, cities, countries, movies, and organizations.



NELL Knowledge Base Browser

CMU Read the Web Project

[log in](#) | [preferences](#) | [help/instr](#)

- religion
- malgorithm
- date
 - year
 - month
 - dateliteral
 - dayofweek
- medicalprocedure
- sociopolitical
- language
- profession
 - jobposition
- mlarea
- color
- creativework
 - book
 - lyrics
 - poem
 - musicalbum
 - movie
 - televisionshow
 - visualartform
 - musicsong
- politicaloffice
- sportsteamposition
- charactertrait
- time
- celltype
- physicalaction

(category)

[View list](#) | [map](#) | [metadata](#)

3,514 instances, 1 page

Any algorithm or class of algorithms that is used at any st

instance	iteration	date learned
clustering as optimization	888	04-dec-2014
conditional random fields and rich feature sets	888	04-dec-2014
cyclic graphical models	888	04-dec-2014
decision trees and boosting methods	888	04-dec-2014
discriminative random fields	888	04-dec-2014
em algorithm for maximum likelihood calculation	888	04-dec-2014
forward backward algorithm	0	28-oct-2010
generalized linear models	1103	18-mar-2018
generalized regression	888	04-dec-2014
hidden markov models	120	19-jun-2010
hidden markov models and viterbi	888	04-dec-2014
hidden markov model hybrids	888	04-dec-2014
hmms	120	19-jun-2010
in dynamic bayesian networks	888	04-dec-2014
kruskal wallis one way analysis	1095	18-jan-2018
latent semantic analysis and pls	888	04-dec-2014
likelihood ratio	888	04-dec-2014
likelihood ratio test	829	10-apr-2014
linear prediction	883	02-nov-2014
linear regression models	104	23-may-2010

startUS
INSIGHTS

277

STARTUPS ANALYZED

5 Top Emerging Knowledge Representation & Reasoning Startups

ELEMENT AI

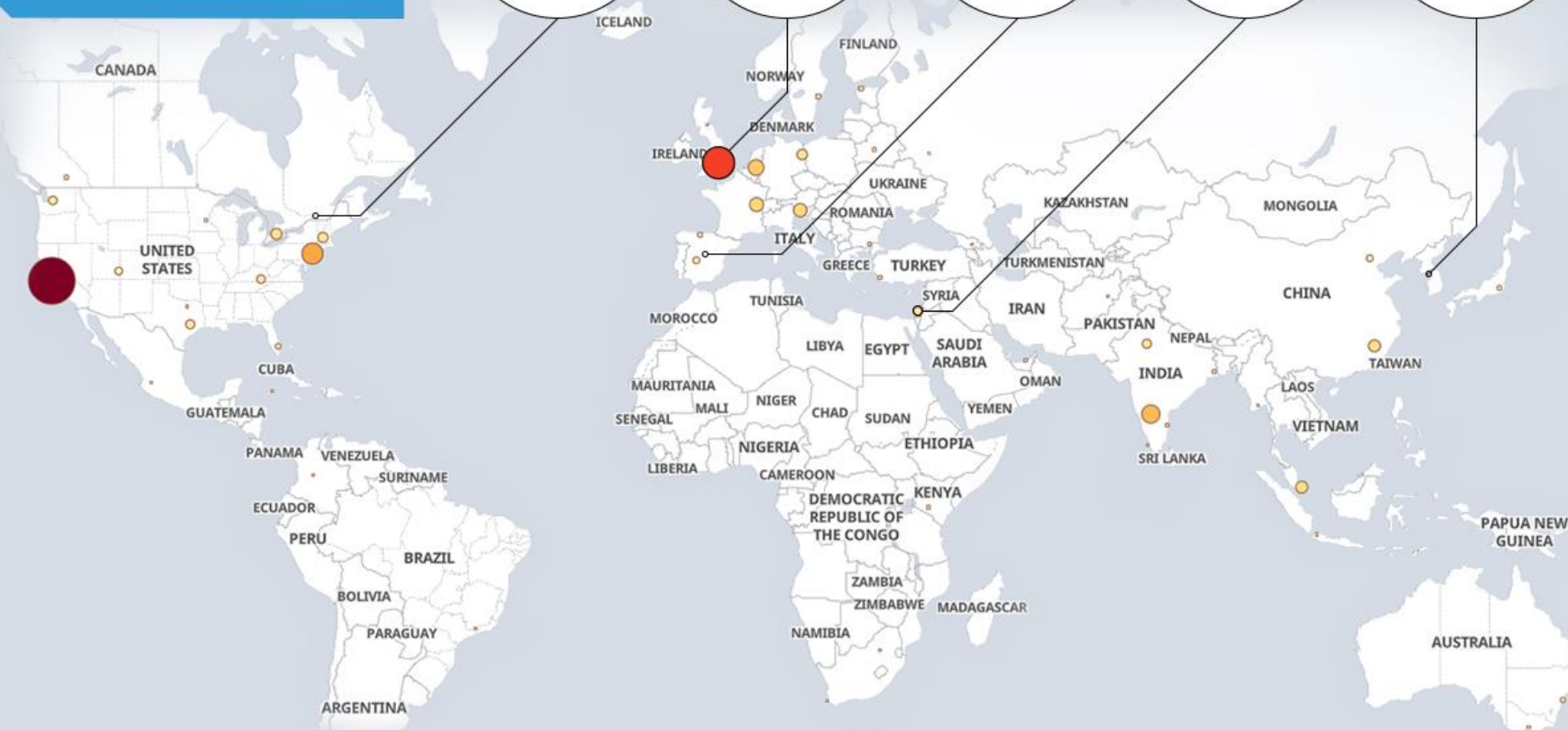
GRAKN LABS

Recognai

timbr.ai



Skelter Labs



This Global Startup Heat Map illustrates geographical distribution of 277 analyzed as well as 5 selected startups. Data from October 2020.